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Workpackage G

Financial transaction data

Deliverable 1: Interim findings for financial transaction data and sharing economy

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Executive summary

This report describes the interim findings of "Workpackage G Financial Statistics". The main aim of WPG is to get an overview of the sources and the data infrastructure (metadata) of financial transaction data in the countries participating in this WP. The objective is to describe to what extent FTD are available as well as whether it is possible for NSIs to access them. Given the infrastructure, it is also a main aim of the WP to assess the statistical potential of these data sources. This may be for improving the existing quality or for quality evaluations of some currently produced statistics, or it may be for a completely new portfolio of statistical products. To do this, empirical studies is being carried out as a stage 2 after knowledge has been obtained on the infrastructure and data have been acquired by the National statistical institute.

In addition to the financial transaction part described above, WPG consists of a part on sharing economy, aiming at utilising financial transaction data to describe the sharing economy.

For the financial transaction data part, this report contains the findings of stage 1, namely, the definition, the main characteristics of payment system and payment instruments, the legal aspect and the main actors moving in this market, besides the description of the financial transaction data situation in the WPG countries: Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. Also included is an assessment of the official statistics that presumably can benefit of using financial transaction data, either as a new data source or as a source acting in addition to the current data source. In this report, this assessment is made prior to accessing data, aiming at suggesting empirical case studies for stage 2. Promising statistics are: household expenditure of residents for tourism and balance of payments estimates, e-commerce turnover, Retail trade index, early estimates of macroeconomic aggregates and indicators, business statistics indicators on flow between industries, and finally turnover of collaborative/sharing economy.

For the sharing economy part, first the theoretical and then the operational definition of sharing economy is discussed. Starting with a "pure" definition limited to sharing of temporarily sharing physical goods between persons, the operational definition adopted is the definition of collaborative or platform economy. The latter is both easier to measure as well as more relevant for economic statistics, enabling e.g. travel accommodation statistics covering more than hotels & regular guesthouses.

After the definition part of sharing economy, different approaches to assessing sharing economy is discussed. The selected approach is to use financial transaction data for finding indicators of sharing economy. A survey is presented, describing the availability of aggregated numbers on credit cards usage found in different WPG countries, and will be used in 2020 for deriving indicators.

The contents of this report will be updated in the final deliverable in the autumn of 2020, where also the results of the empirical studies will be presented as well as the results of deriving sharing economy indicators.

Table of Contents

PART I - Financial transaction data (FTD) ... 8
1. Introduction ... 8
1.1 Terminology ... 9
1.1.1. Abbreviations ... 9
2. Financial transaction data (FTD) (from the finance industry) ... 11
2.1 The different financial transaction data and where to be found ... 11
2.1.1 Payment instruments and platforms ... 11
2.1.2 Processes during the financial transaction ... 12
2.1.2.1 Clearance and settlement ... 12
2.1.2.2 Clearance and settlement Systems ... 13
2.1.2.3 Transfer processing ... 13
2.1.3 The actors Payment service providers and owners of FTD ... 14
2.1.4 Dimensions ... 18
2.2 Getting FTD from an actor having the data ... 19
2.2.1 Legal requirement for getting FTD ... 19
2.2.1.1 Bulgaria ... 19
2.2.1.2 Germany ... 20
2.2.1.3 Italy ... 20
2.2.1.4 Norway ... 21
2.2.1.5 Portugal ... 22
2.2.1.6 Slovenia ... 22
2.2.2 Cooperation process for getting FTD ... 23
2.3 Metadata for FTD ... 24
2.3.1 Population and coverage ... 24
2.3.2 Variables and units: relevance ... 25
2.3.3 More about units ... 26
2.4 Promising official statistics based on FTD ... 26
3. Overview of FTD situation in the WPG countries ... 28
3.1 Bulgaria ... 28
3.2 Germany ... 29
3.3 Italy ... 29
3.4 Norway ... 30
3.5 Portugal ... 31

3.6 Slovenia 31
4. Recommendations 33
PART II - Sharing economy platforms (SEP) 34
5. Definition of Sharing economy and Sharing economy platforms (SEP) 34
5.1 Theoretical definition 34
5.2 Operational definition suitable for official statistics production 36
5.3 More on definition of the platforms 37
6. Sharing economy platform (SEP) data: data accessibility and SEP-related indicators based on FTDs 38
6. 1 Introduction 38
6.1.1 The options developed by the WPG members 38
6.1.2 Option a. The literature review 40
6.1.3 Option b. A non-systematic collection of platforms' indicators 41
6.1.4 Option c. Developing FTD-based SEP indicators 42
6.2 The operationalisation of "option c". 44
The role of platforms in the e-commerce framework 44
6.2.1 The WPG mini-survey on the availability of FTDs to develop SEP indicators 46
6.3 Perspectives for future developments 49
7. Conclusions 52
References 53
Appendices 54
Appendix 1. Terminology related to financial transactions 55
Appendix 2. Bulgaria: financial transaction data (FTD) 58
A2.1 The different FTD 58
A2.2. Getting FTD 59
A2.3 Metadata 59
A2.4 Official statistics potentially benefitting from FTD 59
Appendix 3. Germany: financial transaction data (FTD) 62
A3.1 What transaction data exist? 62
A3.2 Process for getting FTD 62
A3.3 Metadata for FTD 62
A3.4 Official statistics potentially benefitting from FTD 63
Appendix 4. Italy: financial transaction data (FTD) 64
A4.1 Types of FTD 64
A4.1.1 FTD controlled by the Central bank 64
A4.1.2 FTD controlled by private operators 64
A4.2 Getting FTD 64

A4.2.1 The Bank of Italy - Istat collaboration agreement 64
A4.2.1.2 Private operators database 65
A4.3 Metadata for FTD 65
A4.4 Promising official statistics based on FTD 66
Appendix 5. Norway: financial transaction data (FTD) 68
A5.1 The different FTD 68
A5.2 Getting FTD (Process) 70
A5.3 Metadata for FTD 71
A5.4 Official statistics potentially benefitting from FTD 73
A5.4.1 Suggested statistics reported from within the NSI 73
A5.4.2 The promising statistics 73
Appendix 6. Portugal: financial transaction data (FTD) 74
A6.1 The different FTD 74
A6.1.1 Types of FTD 74
A6.1.2 Data owners and data providers 75
A6.2 Getting FTD 75
A6.2.1 Legal requirements for FTD 75
A6.2.2 Process for getting FTD 76
A6.3 Metadata for FTD 77
A6.4 Official statistics potentially benefitting from FTD 77
A6.4.1 Suggested statistics reported from within the NSI 77
A6.4.2 The promising statistics 77
Appendix 7. Slovenia: financial transaction data (FTD) 78
A7.1 The different FTD 78
A7.1.1 FTD controlled by the Central bank 78
A7.1.2 FTD controlled by private operators 80
A7.1.3 FTD controlled by Financial Administration of the Republic of Slovenia 81
A7.2 Getting FTD 82
A7.2.1 Process for getting FTD 82
A7.3 Metadata for FTD 82
A7.4 Official statistics potentially benefitting from FTD 83
A7.4.1 Suggested statistics reported from within the NSI 83
A7.4.2 The promising statistics 84
Appendix 8. What is an operational definition 85
Appendix 9. Questionnaire of the WPG mini-survey on SEP-related indicators 86
Appendix 10. Results of the WPG mini-survey 90

Appendix 11. Problems with sharing platforms 92

PART I - Financial transaction data (FTD)

1. Introduction

Financial transaction data (FTD) potentially contain all financial transactions in a country and can potentially enrich official statistics in various fields such as economic statistics and business structure statistics. In a society where new payment solutions emerge as an option to cash and where the internet enables easier trading, FTD are a natural data source for statistics monitoring the economic activity.

This interim report is Deliverable 1 of ESSnet Big Data work package "WPG Financial transactions". The main aim of the report is to give an interim overview of financial transaction data (FTD) in each of the participating countries in WPG: Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. This inventory involves an overview of types of FTD as well as data infrastructure and metadata of FTD such as population, units and variables. Also, an important question is to what extent FTD are available at all as well as whether it is possible for NSIs to access them. A related question is the ownership of data as well as legal and ethical considerations.

The statistical potential of the FTD sources already being available or potentially being available, is an important part of WPG. Until now within WPG, the FTD have either not been accesses or not been analysed, thus the current report is assessing the statistical potential limited to what insight can be obtained prior to analysing data. This assessment will be updated in the final report, Deliverable 4, after data have been analysed. Then, for the statistics being identified, it will be discussed whether FTD can be used for adjusting the statistics and thus improving its quality, or for quality evaluations of the current statistics. Finally, some FTD might point to a completely new portfolio of statistical products.

One area where FTD could be a useful source for statistics is to measure the *sharing economy*. There is no clear definition of sharing economy beyond temporarily sharing goods, but this will be discussed in Section 5. Given a definition, a subset of FTD consists of sharing economy transactions through sharing economy platforms (SEP). The platforms play a mediating role in connecting the owner of the goods with the consumer trying to access the good. An example is Airbnb for sharing houses/apartments. Measuring sharing economy through FTD is one of the possible approaches to measuring the sharing economy. Another possibility is to address sharing economy platforms directly to get data directly from the platforms. A third and different approach is to perform literature studies that might reveal the metadata of the platform data or the results of empirical studies using platform data.

Section 2 and Section 3 cover the first part of this report, and is an inventory of the infrastructure including metadata, with Section 2 giving a general description on types of FTD (Section 2.1), the legal situation and cooperation with data provider (Section 2.2), the metadata on FTD (Section 2.3), and finally statistics that seem promising for benefitting from FTD. Section 3 provides insight into the special situation in each of the WPG countries Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. Details for each country are presented in Appendix 2-Appendix 7.

In the second part, Section 5 is devoted to defining sharing economy in a suitable way for measuring the impact of sharing economy on economic statistics. Then in Section 6, the different sharing economy definitions are discussed before the three approaches to measuring sharing economy are discussed. The approach of measuring the sharing economy subset of FTD is chosen and operationalised. Finally, a survey of the availability of relevant data in the WPG countries is analysed with the aim of developing indicators of the sharing economy.

1.1 Terminology

A financial transaction is an agreement between a buyer and a seller to exchange goods, services or financial instruments. The two actors are denoted *peers*. The process described here builds up a transaction: 1) a transaction of goods etc. from a *provider* to a *user,* represented by process 1 in Figure 1.1, and 2) a financial transaction going in the opposite direction, represented by process 2 in Figure 1.1.[1]

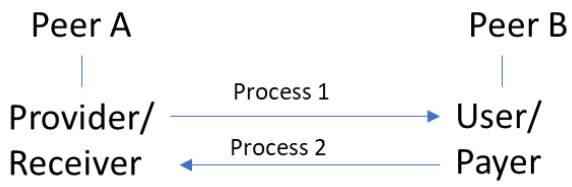


Figure 1.1. The two transactions in a sharing economy transaction.

There are several ways to subdivide and categorise FTD, one of them is to cross-classify into six groups by type of transactions and by type of payer-/payee pair. Transaction types are debit or credit card transactions and "giro"-transactions (bank-to-bank credit or direct debit transfers). Payer-/payee-pair types are consumer-to-consumer (C2C), business-to-business (B2B) or consumer-to-business (B2C)[2]. A consumer in our context can be a person or a household.

As noted above, a peer could be either a business or not a business. In this report, a person is defined as a business whenever the tax authorities define the person as a business. When the peer is not a business, we denote the peer a person, which covers both private persons, households and any other entity not being a business according to the definition above. In the literature it is also common to use the term 'consumer' on the non-business peer, but by using 'person' instead of 'consumer', we can separate the role of being a non-business peer from the role of being consumer of the goods provided in the sharing economy transaction

By *financial transaction data,* we will refer to data on any monetary transaction (between persons, businesses, or between a person and a business) involving the transfer of money between a payer and a payee. Such transactions are also sometimes denoted "payment transactions". Related to financial transaction data are *purchase transaction data*, which consists of receipts and other data about the contents of the purchase but does not contain information about the payer.

A *sharing economy platform* is a business being mediator between the payer and payee in a *sharing economy transaction* where typically under-used goods are shared for a limited time. This will be more precisely defined in Section 13. The sharing economy platform collects/generates data on the sharing transaction. At least some of these *sharing economy platform data are* stored in and/or owned by the sharing economy platform.

1.1.1. Abbreviations

We will use the following abbreviations in the rest of this document:

FTD:	Financial transaction data
SEP:	Sharing economy platforms
P2P or C2C:	Transaction between two persons (sometimes having the role of "consumer")

[1] The payer and the receiver of the process 2, is the user and the provider in process 1, respectively.
[2] For «consumer-to-business», the consumer could be either the payer or the payee.

B2P or B2C:	Transaction between a business and person (independently of the direction, i.e. who is the payer and payee)
B2B:	Transaction between two businesses
NSI	National statistical institute

2. Financial transaction data (FTD) (from the finance industry)

In this section we will look at the common properties of FTD for the WPG-countries. Along with the following country-specific sections this section covers Task 1-3 of the grant agreement.

In the rest of this document, we use terms that might not be widely known outside the banking sector. Therefore, many terms related to financial transactions are described in Appendix 1.

2.1 The different financial transaction data and where to be found

Starting with the perspective of the payer and payee of the FTD, we will first look at the payment instruments and payment platforms. Then we will look at the process behind a financial transaction and the actors within different parts of the process.

2.1.1 Payment instruments and platforms

The payment instruments are:

- Cash, i.e. banknotes and coins.
- Debit cards: Cards enabling cardholders to have their purchases (or cash withdrawals) directly and immediately charged to their bank or payment accounts (except for e-money accounts), whether held with the card issuer or not.
- Delayed debit cards: Cards issued with a contractual agreement granting a credit line, but with an obligation to settle the debt incurred at the end of a pre-defined period without charging the cardholder any interest rate (commonly referred to as "charge cards").
- Credit cards: Cards enabling cardholders to make purchases and/or to withdraw cash up to a pre-arranged ceiling. The credit granted may be settled in full by the end of a specified period or may be settled in part, with the balance taken as extended credit on which interest is usually charged.
- Prepaid cards - a card on which a monetary value can be loaded in advance and stored either on the card itself or on a dedicated account on a computer. Those funds can then be used by the holder to make purchases. Prepaid card is a typical example of e-money.
- Credit transfers ("Giro")- payment instruments based on payment orders or possibly sequences of payment orders made for the purpose of placing funds at the disposal of the payee
- Direct debits ("Giro"[3]) - payment instruments based on preauthorised debits, possibly recurrent, of the payer's account by the payee

In addition, cheque[4] is another payment instrument, but less used today and we will not consider it.

Payment platforms are:

- POS (point of sale) terminal - devices typically used at a retail location to capture payment information electronically and – in some cases – on paper vouchers
- ATM (*Automated teller machines)* - terminals that allow authorised users, typically by using a card, to access a range of services such as cash withdrawals, balance enquiries, transfers

[3] i.e. direct transfer from a bank account to another bank account.

[4] Negotiable payment instruments based on written orders from one part (the drawer) to another (the drawee, normally a bank) requiring the drawee to pay a specific amount from a specified transactional account held in the drawer's name with that institution to the drawee, or a third part specified by the drawer.

of funds and/or acceptance of deposits. Note that not all ATMs need to have cash withdrawal functionality.

- web - electronic transactions conducted via the internet or other computer-mediated (online communication) networks.
- Mobile payment solution - A solution used to initiate payments for which the payments data and the payment instructions are transmitted and/or confirmed via mobile communication and data transmission technology through a mobile device. This category includes digital wallets and other mobile payment solutions used to initiate P2P and/or P2B transactions.
- Traditional mail/telephone - Traditional mail was the traditional platform for paying giro, and about three decades ago, telephone also appeared as a platform for giro. Also, telephone is used for some kind of card transactions (so called "mail order/telephone order").

Traditional mail/telephone is a platform less used today and its use is decreasing. It will be disregarded for the rest of the document.

The payment instruments and platforms can be combined. Table 2.1 shows all the possible combinations: of course, cash payments are not possible on the web and on a mobile payment solution.

Table 2.1. The possible combinations of payment instruments and payment platforms. 'x' indicating a possible combination.

Payment instrument	POS terminal	ATM	Web	Mobile Payment Solution
Cash	x	x		
Prepaid cards	x	x	x	x
Delayed debit Cards and credit cards	x	x	x	x
Debit cards	x	x	x	x
Direct debits ("Giro") [1)]			x	x
Credit transfers ("Giro") [1)]	x	x	x	x

[1)] As a standardised system for transfers between bank accounts, "Giro" can also be considered as a separate payment platform

2.1.2 Processes during the financial transaction

After a payment is initiated by the payer, until the payee has received the data and all accounts have been settled, we can distinguish two types of processes: 1) clearance and settlement, and 2) transaction processing.

2.1.2.1 Clearance and settlement

The retail payment system market generally uses the Clearing and Settlement Mechanism model, in which one or more operators perform *clearing* (i.e. transmission, matching, confirmation of payments and calculation of a final settlement position) and *settlement* (fulfilling the obligations that arose during clearing). For a Clearing and Settlement Mechanism to run smoothly there must be rules governing its operation and participants' access and exit criteria, as well as specific functionalities and technical standards that interconnect all the participants and the other systems. We will briefly describe these mechanisms below.

- Clearance means the calculation of the process of transmitting, reconciling and, in some cases, confirming transfer orders prior to settlement, potentially including the netting of orders and the establishment of final positions for settlement. Settlement is the actual transfer of money between the banks/financial institutions to finance the retail payments / micro transactions. The settlement can be

done “net” or “gross”: a) net settlements: a net position is calculated as a result of the clearance process described above. The net position for transactions happening during the time interval in question is transferred from one entity (the one with the largest sum of payments) to the other (the one with the smallest amount of payments); b) Gross settlements: The amount that shall finance a single micro transaction is transferred from the payer's bank / financial institution to the payee's bank / financial institution. Gross settlements may take place for every single micro transaction or just for some transactions, like transactions exceeding a specific large amount.

2.1.2.2 Clearance and settlement Systems

Below we will describe the clearance and settlement system used in the EU. The Bank of Italy, along with Deutsche Bundesbank and Banque de France, contributed to develop the payment system TARGET2 (T2) which is used within all of EU, and settles the most part of the wholesale transfers, on a gross-basis in real-time. As for the retail payments, the payment system is not fully integrated at the European level yet. However, since 2014 the Single Euro Payments Area (SEPA, onward) has strongly fostered the standardization and interoperability among different national clearing and settlement retail systems, especially for credit transfers and direct debits instruments.

A retail payment system operating across many countries of SEPA is the STEP2-T retail payment system for clearance and settlement. It is run by the private company EBA CLEARING via a pan-European platform, for the centralised settlement of SEPA payments by participating European banks. As a cornerstone of the SEPA processing infrastructure, STEP2-T provides full reach for SEPA Credit Transfers (SCT) and SEPA Direct Debits (SDD) to over 4,800 payment service providers across all SEPA. Payment transactions and retail collections in STEP2-T are forwarded in the form of files to the central system of EBA through the service provider SIA, where they are processed for clearing and transferred for regulation on the TARGET2 platform. In this system, the processing solutions for the interbank management of collection and payment instruments SEPA Credit Transfer (SCT) and Direct Debit (SDD) are offered by SIA and provided in compliance with the SEPA standards in force.

Banks belonging to different banking groups can also exchange retail payments through TARGET2 payment system. T2 (TARGET2) is based on a single shared platform and, from a legal point of view, it is structured as a set of national systems. TARGET2 settles payments on accounts opened by participating banks at their national central bank: therefore, the presence of central bank money makes the system safer compared to using commercial bank money, which is subject to the risk of default by the bank where the settlement account is held. This characteristic enhances the robustness of the platform to financial crises, thereby helping to mitigate systemic risk. TARGET2 is used to settle payments relating to monetary policy operations, interbank payments, operations on behalf of customers of participating banks and ancillary system transactions (i.e. customer payment systems, securities settlement systems, central counterparties and money markets)

For the non-EU country Norway, the clearance and settlement are performed within Norwegian organisations, and a large part of the clearance is done through a net clearance system implying an aggregation of single transactions to the net transfer between the banks during a certain time interval. Details of this clearance and settlement system is described in Section 3 and in Appendix 5.

2.1.2.3 Transfer processing

While clearing and settlement are performed to timely secure and provide liquidity to finance retail payments, every single payment is made with the intention of timely transporting money from the payer to the payee. The value for all statistics produced to measure consumption lies in the information connected to retail payments.

When gross settlements take place, clearing and settlement providers may be useful as a source for information about every payment made from a bank account. Otherwise, payment service providers /actors registering retail payments should be used as a source. Payments with giro, cheques and debit cards take place by debiting a bank account. To find information about payments with these instruments, banks or the service providers processing bank data, will be a natural data source. For information about payments with delayed debit cards, credit cards and prepaid cards/prepaid accounts, the actual issuer of card/account should be used as a data source.

2.1.3 The actors Payment service providers and owners of FTD

The transactions among the economic actors need to rely on a well-designed payment system, that is on a complete set of payment instruments, intermediaries, rules, procedures, processes and interbank fund transfer systems aimed at facilitating the circulation of money in a country or currency area. In this sense, a payment system comprises three main elements:

1. the payment instruments (i.e. payment cards such as debit, prepaid, credit; credit transfers and direct debits), which are a means of authorising and submitting a payment;
2. the processing, which involves the payment instruction being exchanged between the banks (and accounts) concerned;
3. a means of settlement for the relevant banks (i.e. the central bank money).

It also relies on institutions that provide payment accounts, instruments and services to customers and on organizations that operate payment, clearing and settlement services.

The payments between financial institutions are named wholesale payments while those made by banks or other payment service provider on behalf and between non-financial institutions (households, firms, government) are named retail payments.

Data referring to payment card transactions (debit/prepaid or credit), credit transfers and direct debits will usually be owned by banks. In the case where the transaction is processed between two accounts in the same bank, this bank is the owner. When two banks are involved, they own the data together. Thus, for many transactions there is a shared ownership. To handle such transactions between different banks, the financial institutions have established privately owned service providers with the role of operating the transactions. These service providers can take care of the technical role as bank data centrals, and the acquiring as well as the clearance and settlement role. Different entities can have the same role, where each of them is covering different parts of the financial market, e.g. different banks.]

Central banks have a keen interest in the smooth functioning of the national payment system and the way it affects the economy since modern economies are dependent on the safe and efficient flow of transactions. They are involved in payment, clearing and settlement in many different ways: as operators and providers of settlement services in central bank money; as participants in or users of such systems; as oversight/supervision authorities; and as promoters of efficiency in the payment system as a whole.

The central bank usually has a role in settlement, but it may also have a role in the clearance of retail payments. If we look at the role of the central bank in the financial transaction market more broadly, it is somewhat different across countries (as we will see in Sections 7 – 12), but there are also some similarities. The availability of payment data at the central bank depends on the different role of the central bank in the payment system:

- supervision/oversight: All central banks have a role in supervising/auditing the financial market, and this necessitates the acquisition of certain data from the market, that might or might not be relevant for the NSI;

- clearance/settlement role: The national banks have all a common role in contributing to the development of an effective financial market. Still, the central banks have different roles in clearance and settlement of the transactions.

Official statistics production: one difference concerning the role of the national banks, is that national central banks in EU have the responsibility of all the official statistics on the financial market and thus for that purpose they produce payment data in-house. On the contrary NSI's have this responsibility only in some other countries. As an example, in Norway some of the financial market statistics is made by Statistics Norway.

In Section 2.2.1, we will look at the differences in the role of the national central banks from a legal point of view.

Payment data referring to debit cards, credit cards and prepaid cards are owned by every single bank or financial institution that has issued the actual card. The ownership of data connected to the receipt of amounts from these cards will usually be a payee's bank, but this depends on how/to which account the amount is transferred. International payment cards, that can be both debit cards, delayed debit cards and credit cards, are issued by banks under the brand managed by international card schemes. The international card schemes define standards and rules for cards that can be used worldwide such as VISA, MasterCard, Diners, American Express etc. assigning licenses to local banks or financial corporations as principal members for issuing card or acquiring merchants. As the licenses also cover the duty to process transactions, with such licenses banks or national card companies (via their card processors) may be valuable as a data source for card payments information. International card companies also assign roles to local banks or other financial corporations as acquirers of merchants (POS). Since acquirers (through the processor) processes card payments on behalf of a merchant, they may be useful as a data source for information connected to user locations. They will also be able to give information about the geo-localisation of the usage of cards (at national level or abroad)

Payment card transactions and credit transfers may be addressed through mobile devices (mobile payments), and the operational framework can be handled by banks, card operators as well as separate mobile payment operators. In case of "closed loop" payment models, payments between payer and payee are settled in a separated e-money account (prepaid account) such as in the case of PayPal. In case of "closed loop" models, a card (or direct debit) transactions initiated at the bank account can be done for pre-funding the e-money account.

Who has FTD and thus is a potential target for NSIs needing it? In addition to the owners of the FTD, other service providers mentioned above involved in the actual processing of payments have such data, sometimes also national central banks. In some cases, they may have data on behalf of many banks/financial institutions. However, the service providers typically have only the FTD relevant for their role in managing the financial transactions.

The access to FTD might not necessarily have to go through their owners. Both the legal authorisation as well as the technical access can sometimes go through the entities mentioned above instead of accessing through the owners, but this depends on each country's legislation. Since there are many banks in each country, this will often mean more efficiency and less work both for the NSI and for the banks to get direct debits and debit card transactions through a service provider. For international card data, this depends on the number of actual issuers of cards v/s entities with licences as issuers and/or acquirers in the market.

Above, we have the following list of entities having/maintaining FTD in addition to banks:

- National central banks
- interbank clearing/settlement entities

- private payment service providers for banks
- credit / debit / prepaid cards operators
- mobile payment operators
- tax / control authorities' data

Table 2.2 summarises all the FTD data considered within the WPG by each participating country. The table presents on the left-hand side the already accessed data and on the right the possible data to be accessed during the Project. For each country, the data accessed are broken down by type of maintainer (e.g. National central banks, interbank clearing/settlement entities, credit or debits cards operators, tax authorities or other national control authorities). The table shows the complexity of the framework and the actors having card data in each country. Among the already accessed sources, the interbank clearing/settlement entities are slightly prevalent (see the cases of Italy, Norway and Portugal), but also the credit card providers are very relevant data providers for our project (Italy and Norway, and possibly in Bulgaria and Slovenia).

Table 2.2. Framework of the considered FTD data (already accessed or not) in each WPG participating country so far, by type of maintainer.

Country	Types of (accessed) data by type of maintainer				Possible data (not already accessed) by type of maintainer		
	NATIONAL CENTRAL BANK DATA	**INTERBANK CLEARING/SETTLEMENT OF THE SYSTEM OF PAYMENTS DATA**	**CREDIT / DEBIT CARDS OPERATORS DATA, or private payment service providers for banks, or mobile payment operators**	**TAX / CONTROL AUTHORITIES DATA**	**NATIONAL CENTRAL BANK DATA**	**CREDIT / DEBIT CARDS OPERATORS DATA**	**TAX / CONTROL AUTHORITIES DATA**
BULGARIA					**Bulgarian National Central bank** data **-** data from the RING system (established in 1996, it performs real time gross settlement of all payments in the national currency within the territory of Bulgaria) and from another settlement system for interbank payments in euro.	Data from the Company that maintains card payments technical infrastructure	
GERMANY	Total number and value of transaction/payment by type of card and terminal.						
ITALY	**Bank of Italy - UIF Anti-money laundering aggregate reports (SARA) database** of transactions over 15K euro**.**	**Bank of Italy data from the inter-exchange and settlement systems BI-COMP and TARGET2 retail of the System of payments**	**Nexi S.p.A data -** it is a PayTech not only a processing entity but also a credit card company, as well as a primary bank issuing debit card. Information collected both on the issuing and on the acquiring side.			**SIA** (**Società interbancaria per l'automazione) data; Sia is** a processing entity both for issuing and acquiring side (debit and credit card)	
PORTUGAL		**Interbank settlement of the System of payments - SIBS data.** It is connected to the main issuers of payments in Europe, in all operations and in issuers, with several units and several networks - Visa, MasterCard, American Express, UnionPay). Data refer to Data of ATM Express and MULTIBANCO networks, as SIBS send to Statistics Portugal monthly data (only) at NUTS IV level of cash withdrawals and POS payments and they are published by Statistics Portugal by municipality**.**		**Tax and Customs Authority (AT) data** - E-Invoice data E-Invoice is the invoice confirmation system issued under a taxpayer number (communicated by the entity that provides the service or sells the good). The following information will be available to Statistics Portugal: info: VAT Number (taxpayer number),VAT of the buyer, Month of transaction, Country (of the buyer), Taxable value (by VAT Number)	**Bank of Portugal data** - it is manages information on the SEPA (Single Euro Payments Area) financial transfers such as: name of the payer and/or I**BAN of the payer's** payment account; amount to be transferred; IBAN of the payee's payment account		
SLOVENIA						**BANKART data** - it is a company founded by the 22 Slovenian banks processing of ATM operations, card business, payment systems and services (SEPA transactions in both SEPA and the SIMP-PS payment system and the E-invoice system), ATM and POS terminals**.**	**Fiscal cash registers data** - Financial Administration of the Republic of Slovenia monitors all issued invoices, which are paid in cash. This includes all payments except credit transfers, meaning also payments with credit or debit cards are include. adopted measures against grey economy and tax evasion**.**
NORWAY		**Interbank clearing system for the Norwegian Krone (NOK) - NICS data.** NICS manages bank account transfer covering both P2P, P2B,B2P and B2B	**Private provider debit card data**. Data are collected on the acquiring side.				**DSOP data -** Digital cooperation between public and private sectors. Source of B2B,P2P,B2P transactions, the new system will cover both debit card data and bank account to bank account transfers.

2.1.4 Dimensions

We have so far identified three dimensions regarding FTD:

- Dimension 1: Entities having/maintaining FTD
- Dimension 2: Payment instrument
- Dimension 3: Payment platform.

In Table 2.3, we have represented Table 2.2 in a different way by seeing what is the status of getting FTD for the combinations of dimension 1 and dimension 2.

Table 2.3. The existence of FTD in WPG countries, by payment instrument and by type of entity maintaining the data.

Payment instrument	Type of entity maintaining data			
	Central bank	Clearing/settlement	Card operators	Authorities
Cash				SI*
Delayed debit and credit cards	DE	IT*, SI?	IT*, NO***	NO**
Debit cards	BG, DE, IT*	IT*, SI	IT*, NO	NO**
"Giro"	BG, IT*, PT?	NO*, SI, PT		NO**, PT?

* test data are already accessed or will soon probably be available for the NSI (covering part of the population)
** will appear as a data base outside NSI in a couple of years.
*** Such data have been identified also in Norway, held by the Issuers.

Let us consider expanding Table 2.3 by also including payment platform, the third dimension. The question then is: is data from a payment instrument handled differently by the actors of Table 2.3, depending on whether payment was done (at POS terminal or on the Web)? In Norway, as an example, one of the largest providers of payment systems data has information on all the use of card data in POS terminals accepting the national card scheme. To receive information about use of cards in an ATM or over the internet, we have to go to the owner of the ATM (for ATM transactions) or card issuer or the entity with the issuing license (for both ATM transactions and transactions over the internet).

As we have seen both in Table 2.1 and Table 2.3, there are some pairwise combinations of the three dimensions that will not exist. We can see which combinations of all three dimensions not being possible in Table 2.4. All or some of the cells of Table 2.4 will be needed to cover as much as possible of the statistical population for certain statistics to be produced. Notice that for a given row (payment instrument), a data maintainer might contain all POS-data and all web-data, and in that case only the corresponding columns are relevant for this statistics. This also means that the relevant cells of Table 2.4 for covering as much as possible of the statistical populations, will sometimes be different depending on the country.

Table 2.4. Types of FTD by payment instrument, by type of entity maintaining the data, and by payment platform. Grey cells indicating impossible combinations

	Type of entity maintaining data							
	Central bank		Clearing/ settlement		Card operators		Authorities	
Payment instrument	POS/ ATM	Web	POS/ ATM	Web	POS/ ATM	Web	POS/ ATM	Web
Cash								
Credit cards								
Debit cards								
"Giro"								

2.2 Getting FTD from an actor having the data

To get data from the different actors (cf. Section 2.1.3), two essential activities to implement are: 1) meeting national legal requirements, and 2) setting cooperation agreements.

2.2.1 Legal requirement for getting FTD

In considering the legal aspects allowing or facilitating the access to the FTD we must consider simultaneously two dimensions, the prerogatives that the national statistical law (in the provided forms) guarantees to the National statistical institutes (NSIs) for statistical production purposes and the prerogatives of autonomy in the field of the supervision and the control of payment system that the law provides to the central banks in the different countries.

Among the countries participating in the WPG on FTD of the ESSnet Project BIG DATA II, there are similarities regarding the legal requirement ruling the general data access set by the national statistical acts, but major differences in the norms setting the relationships between the NSI and the National central banks. This is due to fact that four participating countries belong to the European system of central banks (ESCB) (Germany, Italy, Portugal and Slovenia) whereas Bulgaria and Norway do not belong to the Euro area and consequently present different rules.

In line with art. 105 of the Treaty establishing the European Community and in line with art. 3 and 22 of the Statute of the European System of Central Banks and the European Central Bank, individual national central banks of the Eurosystem may provide processing facilities for retail payments in euro for credit institutions, either via participation in private retail payment systems or acting as operators of their own retail payment systems, in order to contribute to the safety and efficiency of payment systems in the euro area. Depending on the specific national circumstances, they may also facilitate access to payment systems for all credit institutions.

Regarding the general principles ruling the data production in the ESCB, the Article 5 of the Protocol on the Statute of the European System of Central Banks and of the European Central Bank *(Official Journal of the European Union C 202/230, 7.6.2016)* sets for the ESCB, assisted by the national central banks, the tasks of data collection of the necessary statistical information for its aims. The national central banks carry out the collection, and the ESCB contribute to the harmonisation of the rules and practices governing the collection, compilation and distribution of statistics. The ESCB attaches great importance to the quality of its statistics. It therefore takes into consideration internationally agreed quality standards, such as those formulated in the IMF's Special Data Dissemination Standard and Data Quality Assessment Framework, which are in turn rooted in the UN's Fundamental Principles of Official Statistics.

Nevertheless, without prejudice to the cited Protocol, the ESCB collaborates with the European Statistical System (ESS), which comprises Eurostat, the National statistical institutes and other national statistical authorities, and takes account of the principles laid down in the European Statistics Code of Practice for the National and Community Statistical Authorities.
The Article 21 of EC Regulation n. 223/2009 and the Article 8 of the EC Regulation n. 2533/98 establish that the exchange of confidential data between ESS and ESCB subjects is authorised if necessary for the efficient development, production and dissemination of European statistics or improvement of their quality and that this need has been justified.

An outline of the main legal features at national level may highlights the national specificities.

2.2.1.1 Bulgaria

Statistical Act stipulates "The National Statistical Institute shall collaborate with the Bulgarian National Bank, including by means of exchange of individual data for statistical purposes, in developing, producing and

disseminating official national or European statistical information or the statistical information of the European Central Bank." Art.4 (5)

Further on Art.20 (6) says "Central and local government authorities, other central government institutions, the Bulgarian National Bank and the bodies keeping registers and information systems as stipulated by a law, shall have the obligation, upon written request and free of charge, to provide the National Statistical Institute and statistical authorities with the individual data collected by them and/or statistical information that is necessary for conducting the statistical surveys included in the National Statistical Programme".

2.2.1.2 Germany

The activities of the Federal Statistical Office in Germany are based on the Act on Statistics for Federal Purposes (Federal Statistics Act - BStatG) in the version promulgated on 20 October 2016 (Last amended by Article 10 paragraph (5) of the Act of 30 October 2017 and published in the Federal Law Gazette I, p. 3618). The act is originally published in German language. The provisions cited here are based on an English working draft. But only the German version is authentic.

According to the Federal Statistics Act every statistical survey has to be based on a legal regulation. Apart from the case of an European regulation this can be an act adopted by the legislative or another form of legal regulation enacted by the government (Stipulated in Section 5 of the Federal Statistics Act). Further provisions (Sections 5a, 6 and 7 of the Federal Statistics Act) give attention to:

- The use of administrative data
- Measures for the Preparation and Production of Federal Statistics and
- Surveys for Special Purposes.

Altogether, the activities of the Federal Statistical Office are not determined by itself but by the government and the ministries respectively.

In Section 3 of the Federal Statistics Act concerning the duties of the Federal Statistical Office the relationship to the national central bank (Deutsche Bundesbank) is described: "[The Federal Statistical Office] shall cooperate closely with the Deutsche Bundesbank to minimise the effort involved in data collection and to ensure the quality and coherence of the statistics compiled."

The tasks of the national central bank are stipulated in the Act on the Deutsche Bundesbank (Bundesbankgesetz – BBankG). Section 18 of this act stipulates that the Deutsche Bundesbank is authorised to arrange and to conduct statistical surveys in the field of banking and finance if this is necessary to perform its tasks.

In addition to these provisions stipulated by law, the Federal Statistical Office and the Deutsche Bundesbank have in 2014 agreed to a common Memorandum of Understanding. In this Memorandum the cooperation between both institutions is concretised. Especially, it is stated that both partners provide each other with statistical data concerning the fields of activities that are subject of their cooperation.

2.2.1.3 Italy

The statistical legislation regarding the NSS is mainly described in the Italian Legislative Decree no. 322 of the 6 September 1989 (and subsequent amendments) and in the Annex A3 to the Privacy Code (the Code of Ethics and good practices for the NSS). According to that legislation, Istat is in charge of proposing a National Statistical Plan (NSP) in which it establishes statistical projects representing a "substantial public interest". The NSP is submitted to the Committee for Safeguarding Statistical Information for advice and then approved and implemented by government's decree. The NSP establishes, among others, the purposes, the appropriate safeguards, the information notice and for which specific projects participation of natural and

legal persons must be deemed as compulsory. Public and private entities can consequently be requested by the NSS to provide individual data in the context of such a mandatory project.

The processing of personal data must also respect the provisions of the Code, Annex A3 and the Statistical Legislation. All projects whose participation is not explicitly defined as mandatory in the NSP are deemed as voluntary and are submitted to the application of the usual rules of the Code and Annex A3 if personal data are processed. Regarding the secondary use of personal data for statistical purposes, it is always possible to use for statistical purposes personal data that third parties have collected for whatever purpose. Nevertheless, Istat has the obligation to inform private and public subjects, which hold or disseminate data through the various multimedia channels, about the specific uses of the processed data.

In carrying out its institutional functions, the Bank of Italy collects data and produces statistics on money, credit, balance of payments and other economic indicators. According to the 1993 Banking Law (art. 146), the legal provisions of the Treaty on European Union (art. 127.2) and the Statute of the ESCB (art. 22), Bank of Italy as oversight authority[5] of payment systems collects payment transactions data from payment systems, bank and non-banks operators and produce (aggregated) statistics on payments.

Moreover, pursuant to the Italian Anti-Money Laundering Law (Legislative Decree no. 231/2007), banks and other financial intermediaries have to report periodically to the Financial Intelligence Unit established at the Bank of Italy (UIF) payment transactions over certain thresholds, after aggregating them by branch, customer sector and type of transaction.

Payment data as well as other data collected within the supervision duties of the Bank (i.e. payment system oversight, bank supervision, anti-money laundering controls) are covered by secrecy and confidentiality rules. However, some data may be available for official publication in aggregated way.

2.2.1.4 Norway[6]

Norway is neither a member of EU nor of the ESCB. The national central bank has a similar role as those in EU when it comes to supervising and auditing the financial market, but the role is different when it comes to official statistics production and dissemination. The Central bank of Norway disseminates the statistics on the financial market. However, contrary to the ESCB system, the financial market statistics of Norway have been produced by Statistics Norway since the Central bank unit that used to produce this statistics, moved to Statistics Norway in 2007. This also means that the legal authorisation for getting the FTD data was changed. The central bank get data from the Central bank Act §27 when needed to perform its main tasks but also for production of Official statistics.

The Statistical Act §2-2 grants Statistics Norway access to any register that Statistics Norway needs for production of official statistics, unless access is prohibited by another law. This right is granted whether the register is governmentally owned or privately owned. When Statistics Norway needs data stored in a register for statistical production, a decision is made which requires the register owner or another governmental or private organisation/business having the data to provide the register. There is an exception when the organisation is just a data handler. Anyhow, a good cooperation process with the data provider (whether being owner or not) is important also in a situation with so strong legislation.

[5] With this regard, according to the banking law and the Eurosystem framework Bank of Italy within the role of payment system overseer and promotes the smooth functioning of the payments system through its direct management of the main circuits and by exercising oversight powers of guidance, regulation and control. This activity, coupled with market supervision, is intended more generally to contribute to the stability of the financial system and foster effective monetary policy.

[6] The description is based on the legal acts operating in 2019.

The paragraph above applies to all registers. The question is then whether access by Statistics Norway is prohibited by other acts. For FTD, the "Financial Institutions Act" and "Central Bank Act" are relevant.

The Financial Institutions Act, §16-2 (1) is about obligation of secrecy and implies that restrictions are not preventing access authorised by another legal act. As seen above, the Statistical Act authorises such access if these is a justified need due to official statistics production.

Obligation of secrecy is covered by the Central Bank Act §12 and prevents access by others with some exceptions. One of the exceptions is that the Public Administration Act §13b no. 4 applies. Together with the Statistical Act §2-2, this authorises Statistics Norway access to Central bank data (given a justified need due to official statistics production).

2.2.1.5 Portugal

The National Statistical System (NSS) includes the Statistical Council, the State body that superintends and coordinates the system; the National Statistical Institute (Statistics Portugal), the central body responsible for the production and dissemination of official statistics, that ensures the supervision and the technical and scientific coordination of the NSS; the Bank of Portugal that, as part of its mission, is responsible for the collection and compilation of monetary, financial, foreign exchange and balance of payments statistics; the Regional Services of Statistics of the Autonomous Regions of Açores and Madeira that act as delegations of the Statistics Portugal in relation to nationwide statistics as statistical authorities in what concerns regional statistics; and other entities producing official statistics by delegation of Statistics Portugal. Statistics Portugal, Bank of Portugal, Regional Services of Statistics of the Autonomous Regions of Açores and Madeira and entities producing official statistics by delegation of Statistics Portugal are considered statistical authorities, having responsibility for the production of official statistics, and empowered to require (mandatory and gratuitously) to all departments or agencies, individuals and legal entities, information necessary for the production of official statistics.

The Bank of Portugal collects and compiles monetary, financial, foreign exchange and balance of payments statistics, particularly within the scope of its cooperation with the European Central Bank (ECB) (Article 13 of the Organic Law). The Law on the National Statistical System (NSS) (Law No 22/2008 of 13 May 2008) acknowledges the Bank of Portugal as statistical authority without prejudice to the guarantees of independence deriving from its participation in the European System of Central Banks (ESCB). According to SEPA (Single Euro Payments Area), the Bank of Portugal is able to manage several information on the financial transfers, such as the name of the payer and/or IBAN of the payer's payment account; the amount to be transferred; the IBAN of the payee's payment account.

2.2.1.6 Slovenia

According to the Statistical Act (ZDSta), reporting units that are holders of official and other administrative databases are by the statistical survey program defined as data providers (Article 4). According to Article 35 of the National Statistics Act, these reporting units must provide the Statistical Office and the authorised performers with complete and correct data free of charge, in a timely and in a prescribed manner. This duty lasts as long as they perform the observed activity or until the reporting obligations cease. When compiling statistics for which no compulsory reporting is specified, reporting is based on the agreement of the reporting unit (at its sole discretion and on a voluntary basis). According to Article 5 of the National Statistics Act, the reporting unit must be acquainted with the basic characteristics of the statistical survey, with a definite period of statistical processing, as well as with the right to decline cooperation for any reason and at any stage of the statistical survey.

The Banking Act regulates the conditions for the establishment, operation and regular termination of credit institutions established in the Republic of Slovenia and the competent authorities, measures and powers to

exercise supervision over their business. Article 125 of the Act defines that confidential information under this Act is all information, facts and circumstances regarding the individual client at the disposal of the bank. By the Act bank must protect the confidential information referred to in the Article 125, irrespective of the manner in which it was obtained. Article 126 defines cases in which the bank can share data with external users. It is precisely defined to whom the bank can provide information: the Bank of Slovenia, the European Central Bank, the supervisory authorities, the Commission for the Prevention of Corruption, the Commission of Inquiry (in the case of a parliamentary inquiry), the Court of Auditors of the Republic of Slovenia and members of the information exchange system customer ratings. Statistics Office is not defined as one of those users and this represents the biggest obstacle in obtaining the data, both from banks and from private operators owned by the banks.

2.2.2 Cooperation process for getting FTD

A good cooperation process for getting FTD is a crucial activity. It can be regarded as a process following different phases:

- Identifying a need for data
- Collecting overall infrastructure information on the financial market and its actors, both in the micro transaction processing stage, in the clearance stage and in the settlement stage of the transaction processes.
- Identifying the default potential data provider
- Setting clearly the cooperation agreement terms
- Maintaining a good cooperation relation with the potential data provider.

The National central Bank should probably be the first institution that should be addressed to receive a thorough insight into the infrastructure of FTD in a country, in the sense of

- which are, if any, the other governmental institutions involved in the financial market, either by monitoring or regulating the financial market, or playing other roles.
- which are the financial institutions/companies that are involved in financial transactions, both banks, payment card companies and other data providers.
- which are the organisations involved either in monitoring the financial market, being "spokespersons" for the actors in the financial market etc.

The National central Bank accesses FTD as a part of their auditing role and may or may not be able to provide data for the NSI (cf. Section 2.2.1 on legal requirements at national level).

The private institutions having a role in payment processing, clearance or settlement of the transactions (c.f. Section 2.1.1) are possible data providers. Getting bank-to-bank transfers or card data from the owner means contacting a lot of banks and credit card companies, and this approach is not suggested if there are other possibilities. Alternatives could therefore be processing companies and acquirers, as well as payment card issuing companies.

Sometimes there are several potential data providers, but only one of them is the "default" provider, given its more comprehensive coverage of the population, its infrastructure being more well-adapted to data delivery, or its role fits better with the data provider role than is the case for the other potential data providers. In that case, this default provider should be addressed first, since other potential providers could be negative to provide data when they are not themselves the default data provider. Since data delivery requires quite some resources, it should be based on a clear need of getting the data from *this* potential data provider.

A good cooperation process with a potential data provider is an advantage for the following several reasons:

- The potential data provider has extensive knowledge of its data and metadata, whereas the NSI might not exactly know what to ask for, at least for a first test data delivery. A successful outcome of a data delivery is depending on the data provider to use additional resources and together with the NSI derive what data are needed by the NSI and what metadata should be provided. Inevitably, this process is depending on a constructive and positive cooperative relation between the potential data provider and the NSI.
- Since FTD are generally considered sensitive due to the object (money) being transferred, it is important that the data provider/data owner is confident that the NSI will treat their data with extreme caution and can guarantee a confidential treatment. When data about individuals are collected, data must be treated according to the GDPR rules. Note that providers of card data have a sharp focus on that card data must be treated according to PCI security standards https://www.pcisecuritystandards.org/).
    - A cornerstone in the process of guaranteeing confidential treatment, is to explain the NSI's role to the data provider/data owner. In contrast to most other governmental institutions, the NSI has a sole statistical purpose for its use of the data and will not look at the data for any identified person/local unit/enterprise. The latter will however be a purpose for most of other governmental institutions as a part of their control/surveillance/monitoring activity. It is of importance to emphasise and convince data providers that their data will be treated confidentially, totally safe and for statistical purposes only.
- Although the legal basis for data acquisition is present, the data provider can choose to cause a lot of extra work for the NSI as well as delay through a legal dispute process which not only costs a lot of time and money for the NSI, but also could be negative for the NSI's reputation.

Thus, the first recommendation is formulating a clear need for data as well as an assurance that the data will be adequately secured, but also use of other means for improving the cooperation relation is recommended as the participation in investment costs in developing a reporting framework. Some resources are often required by the data provider to perform a successful data delivery, but the NSI in some countries cannot pay for getting data. However, the NSI can contribute to the cost reduction helping the data provider by letting one of their own employees work inside the data provider's office with the data preparation for a period of time. This is only possible if such an arrangement can be fitted into the confidentially rules of the potential data provider, and at least requires the NSI employee to sign a confidentiality declaration concerning his/her work inside the potential data provider's office, and perhaps also to be employed at the potential data provider for a limited time period, where the salary is paid by the NSI. Notice that the expenses for the NSI not only contributes to getting the data, but also to improve knowledge of metadata inside the NSI.

2.3 Metadata for FTD

2.3.1 Population and coverage

The FTD has as base unit the single financial transaction between the payer and the payee of money. The total of all possible FTD is the set of all base units where the transaction is done electronically. Our population of interest, is however often the "all payments"-population, which consists of all possible FTD together with all payments made in cash, i.e. by coins or banknotes.

Notice that a single credit card transaction is a transaction between the card holder financed by a credit card company, representing the issuer, and the payee. Thus, since a bank account is not debited and the banks therefore are not directly involved in the single transactions, credit card data typically must be received from other sources than when debit card data and other payment instruments for bank to bank transactions (giro) are used.

The coverage of data covering only payment instruments is over time susceptible to changes in the consumer behaviour. It is possible in some countries to receive bonus points for e.g. flights when using certain credit cards. Such incentives could lead to an increased proportion of credit cards usage. The increase of mobile payments may reduce volumes of transactions processed by banks and card companies and increase the volume of transactions processed by mobile payment operators.

2.3.2 Variables and units: relevance

Just as for an ordinary administrative register, FTD are collected for administrative purposes, not for statistical purposes. Below are some challenges concerning metadata, some of them are common to any administrative register whereas others are particular for FTD as well as for other big data resembling administrative registers.

- Neither the units nor the variables are necessarily adapted to the statistical purpose. As an example, for debit card data, the main administrative purpose is to secure that the correct amount of money is transferred between the correct accounts, accompanied by some explanatory details needed for the payer and the payee. In addition, the bank(s) needs the data on a single financial transaction to keep track of the remaining sum of money in the two accounts in question. These needs will influence the units and the variables, and hopefully the units and variables will be adapted to the statistical purpose as well, but as an example the system might not be concerned with whether the bank account is directly connected to a legal unit or a local unit. Somewhere deep in the database system, there will most likely always be a link to a legal unit, but this information might not be easily available for the NSI.
- The infrastructure of the FTD data might not be immediately compatible with the statistical need of representing all the transactions (within a domain) as a dataset (table) containing all the relevant variables existing in the Data provider's system. The main requirement on the infrastructure is to provide each account owner with a table of the transactions related to this account during a period of time, in addition to keeping track of the total amount of money in the account at any time. Furthermore, since FTD contains a huge amount of data, there could be technical/practical constraints on how data can be stored and otherwise handled by the banks and the data provider. Construction of a relevant data set for the NSI could thus be hard and time-consuming work for the Data provider, and some variables could be more difficult to find than others.
- The variable-descriptions received by the NSI could be:
    - Totally or partially missing if
        - the data provider has only poor variable-descriptions, or the metadata system and thus the variable-descriptions are fragmented. For a data provider working only with a limited set of variables being well-known to them, the need for good variable descriptions might not have been acknowledged in the data provider organisation. Instead, everyone could just have the descriptions in their mind, and there is no perceived need of having the written documentation easily available. In such situations, the data provider will have to use much time for producing a list of variable-descriptions.
        - the NSI has taken the variable-description for granted without explaining the NSI's need for such a description when the delivery agreement has been made.
        - the NSI receives additional variables to those specified in the delivery agreement, which was not thought of in the delivery agreement, but which are regarded as

necessary additional variables by technical staff at the data provider's organisation, and that might – or might not - turn out to be useful for utilising the main variables.[7]

- incomplete in the sense of using concepts that are imprecise for a statistical purpose but could very well be sufficient and even precise for the administrative purpose.

2.3.3 More about units

The single transaction can be regarded as a base unit. For a certain official statistics, the transaction might however not be the *statistical* unit. Instead, the statistical unit could be e.g. payer (in Consumer expenditure statistics). Payer is a compound unit with respect to the base unit above, i.e. an aggregation of base units. Using a compound unit, leads to a coarser granularity. The typical unit in the data set received by the NSI will typically be a compound unit. As an example, the unit could be the set of retailers in a region receiving payment(s) related to a certain industry classification code during one day. Then, the transaction amount received by this compound unit is the sum of all transactions received within this region and industry classification code for this day.

In the retail sales example above, the focus for defining the compound unit is the businesses receiving payment. On the other hand, with the consumer perspective, e.g. the consumer expenditure survey, a compound unit based on the same industry classification code and day, could be defined as the set of persons in this region who has spent money this day related to the industry classification code in question. Then, the corresponding transaction amount sent by this compound unit (to different retailers inside or outside this region) is the sum of all transactions sent from persons within this region and industry classification code during this day.

2.4 Promising official statistics based on FTD

Based on FTD, there could be potential for improving current official statistics or for developing new official statistics.

An improvement of current official statistics can be done in three different ways:

- replace the current source: This option has the advantage of reducing the response burden connected with the current data collection that usually is based on industries or persons filling in questionnaires.
  - An example is the retail sales index. Today, shops need to report their total monthly sales to the NSI. FTD aggregated to shop, could potentially replace the current statistics once both credit and debit card data are included in FTD, and in addition the cash payment amount is reduced. In Norway, the cash payment rate has already decreased to about 20%, and it will decrease further in the future and eventually disappear someday. In the future, also giro payments need to be added to the FTD, since internet payment is increasing (web shops where one can order and pay before picking up in the physical shop, or receive by mail delivery).
- Improve estimation based on the current data collection:
  - Still using the retail sales statistics example, FTD will give totals for the whole population of shops, thus the survey-based estimation can be adjusted/calibrated to the totals known through the FTD. If the debit sales total is known in both the sample and the population, then we could use this in a rate estimator, as an example. Such an adjustment can reduce the sample variance as well as non-response and measurement error.
- Quality assessment of the current official statistics:
  - In the retail sales statistics example, there could be reasons not to change the estimator, e.g. due to uncertainty about the FTD coverage or other considerations. Thus, in this setting we

[7] Obvoiusly, these additional variables would not have been received by the NSI unless the data provider has classified it as non-sensitive.

regard the FTD source as having insufficient quality for being included in the official statistics estimator. However, FTD could still be used to assess the quality of the current official statistics. One way of doing this is to produce an alternative estimator as in the bullet point above, and although the alternative estimator is considered worse than the traditional estimator, it can still tell something about the quality of the current estimator.

Before accessing FTD and testing their potential for improving a certain official statistics closer, it is difficult to indicate which official statistic product could benefit the most from the use of FTD and, consequently, on which product is most promising to invest. Thus, it is possible to make such an assessment considering only the information coming from the available metadata at this stage. Every WPG participating country is supported in doing this assessment by the results of an internal survey conducted within the NSIs. The results of these surveys are synthetized in national case studies reported in the Appendices. According to these results, promising official statistics that could benefit from the use of FTD are:

- *household expenditure of residents for tourism and balance of payments estimates* – new data can be used for comparison with official statistics in order to improve the quality of current statistics, as indicated by Italy;
- *e-commerce turnover* - private credit card operators data can be compared to the current e-commerce turnover index (traditionally calculated with surveys) and can help in getting some insights even by MCCs groups, as reported by Italy;
- *retail trade index* – FTD is a potential alternative source to be used either in the estimation of the index or in the quality assessment of the current statistics, either in form of card use data as reported by Norway, or by using the fiscal cash register as reported by Slovenia accuracy;
- *early estimates of macroeconomic aggregates and indicators* – FTD aggregated (monthly or daily) series can be used for nowcasting/forecasting purposes. The availability of private credit card operators FTD will allow refining the forecasting, for instance, of private consumption and retail trade index at the national level or for the early estimates of turnover in Services sector, as proposed by Italy;
- *business statistics indicators* – new indicators on monetary flows among companies can describe the economic relations between industries, as indicated by Norway;
- *turnover of Collaborative economy or Sharing economy* - Portugal indicated that the new FTD sources can help in the production of some indicators measuring these new economic activities dimensions.

Some of these promising statistics are candidates for further empirical studies within the WPG activities, after verifying the feasibility of these studies based on actual FTD-data obtained by private operators and/or governmental organisations. After the evaluation of the-FTD quality during the empirical studies, the list above will be updated in the final WPG report in the autumn of 2020.

3. Overview of FTD situation in the WPG countries

The FTD situation varies between countries. Below is a short overview of the situation in the different WPG countries. More details for each of the countries are presented in Appendix 2 to Appendix 7.

3.1 Bulgaria

Types of FTD

Bulgarian National Bank publishes monetary survey on the monthly base. Monetary survey provides data about foreign assets, money in circulation, deposits and monetary aggregates M1, M2 and M3.

In addition to monetary survey, BNB publishes data about POS payments and online payments. Data are aggregates and do not make difference between legal persons payments and natural persons payments.

Also, BNB publishes more detailed data about deposits of households and Non for-Profit Institution Servicing Households and deposits of non-financial enterprises grouped by amount of deposits. BNB publishes data about credits by amount and industry as well.

Process for getting FTD

Bulgarian National bank, being administrator of settlement system, is a potential provider of granular data on consumers to business payments and business to business payments. Another potential provider is the private company Borica JSC. Borica is a technology company that provides payment services. The company developed and maintained payment platform which is used by its clients.

Commercial banks are another potential data provider. Top five banks hold about 60 percent of overall bank assets in Bulgaria. It is reasonable to focus on the biggest banks when negotiate data access.

Metadata for FTD

Bulgarian National Bank publishes monetary survey each month. Data about deposits grouped by number and value and credits grouped by value and economic activity are published quarterly.

Also, Bulgarian national bank publishes number and value of POS payments and online payments annually. These are aggregated data about consumer to business and business to business data. Consolidation ofB2B and C2B payments is a shortage of the data for we cannot separate final demand related payments from intermediate consumption related payments.

Granular data access is subject to negotiation.

Promising statistics

FTD are much more valuable when combined with national account data and business statistic data. For example, mortgage loans when combined with data about houses built and building permission are much more informative about housing market than considered alone. Consumption loans are more informative when combined with private consumption data and income data than considered separately.

The examples suggest FTD could be very useful when combined with existing data. FTD refer to the money flow within the economy. Business statistics and national accounts refers to the flow of goods and services within the economy. Having data about both good/service flows and money flows could enrich economic statistics.

3.2 Germany

Types of FTD

The Deutsche Bundesbank publishes data about the number and volume of card transactions and payments to POS terminals. The available data are however only at the aggregate level and do not inform about the nature of purchase done per transaction or payment.

Process for getting FTD

The Deutsche Bundesbank is part of the TARGET2 system: About half of the payments in terms of volume and over one-third in terms of value are submitted via the Deutsche Bundesbank through the Target2 payment system, with the German system called TARGET2-Bbk.

Metadata for FTD

In addition to the data passing through the TARGET2 payment system, the Deutsche Bundesbank currently receives monthly data on credit and debit card turnover in relation to travel (balance of payments) including payments cards transactions of German residents travelling abroad as well as payments cards transactions of foreign travellers in Germany. However, more detailed information, as for example the Merchant Category Code (MCC), is not known. The current data therefore does not allow for empirical analyses on estimating shared economy transactions.

Promising statistics

Given the current data available at the NSI, it is not possible for now to identify potential official statistics that would benefit from FTD.

3.3 Italy

Types of FTD

The Bank of Italy directly manages the information on the interbank retail payment systems, through BI-COMP clearance and TARGET2 settlement systems. The EBA-clearance system operated by SIA and NEXI (the main card processors in Italy) is in addition to the national central bank system. The private payment data processing entities manage the data on behalf of banks.

Process for getting FTD

The Italian Institute of Statistics has undertaken a collaboration agreement with the Bank of Italy for the exchange of aggregated statistical information. SIA and NEXI were contacted by the Bank of Italy for an experimental project for sharing aggregated daily information. Data collection from card processors (*SIA and NEXI)* requires an intense experimental phase before proceeding with their exploitation for statistical production purposes, because problems related to either the harmonisation of different IT platforms or the set-up of data lakes arise.

Metadata for FTD

FTD managed by Bank of Italy are timely, with a good broken down by payment instruments but not granular in terms of sectoral and geographical breakdowns, because they come from predefined interbank procedures at the settlement level. We could have access to private operators daily data with information on the location

of the merchant and the issuer, both from the acquiring side and the issuing side (useful for measuring the cross-border flows). Other information allow the distinction between e-commerce and proximity payments (useful to measure the retail trade turnover) and the Merchant Category Code (MCC).

Promising statistics

Promising statistics are 1) *household expenditure of residents for tourism and balance of payments estimates* for comparison with official statistics; 2) *Nowcasting/forecasting of macroeconomic aggregates* - BI-COMP and TARGET2 and private credit card operators series used for refining the forecasting of private consumption, retail trade index, or service sector turnover; 3) *e-commerce turnover by MCCs groups* - private credit card operators used to make comparison with the current e-commerce turnover index and to get some insights by MCCs groups.

3.4 Norway

Types of FTD

- In Norway, there is a combination of a net and a gross clearance and settlement system. The net system for each period (of some hours during a day) contains the aggregated net transactions of transactions between bank A and bank B
- Norway has, in the same way as countries like Denmark and Canada, a national debit card scheme that covers most of the debit card transactions.
- The Register of Crossborder Currency Transactions and Currency Exchange contains all transactions crossing the national border.
- The DSOP control system is a system for cooperation between some governmental organisations and the private financial institutions, making the current governmental audit requests more automatic. The system will start during 2020 and is based on an API interface.

Process for getting FTD

- Clearance and settlement data is not a relevant source for Statistics Norway due to the net clearance system.
- Norway has received micro-transactions of national debit card scheme data for one day in 2016.
- The DSOP control system are being addressed by Statistics Norway for cooperation on future data delivery to Norway. Some keywords are: getting and discussing metadata, preparing for the API solution, and getting synthetic test data.

Metadata for FTD

- The national debit card scheme data covers more than 75 percent of the debit card payments on point of sale terminals.
- The Register of Crossborder Currency Transactions and Currency Exchange: for smaller transactions, the receiver is aggregated to country and month.
- DSOP control will cover all debit card- and giro-transactions. Only samples, e.g. random sample, of persons and businesses will be available for Statistics Norway due to the server capacity limitations within the DSOP control system.

Promising statistics

- Retail sales index

- Economic relations between business industries

3.5 Portugal

Types of FTD

In Portugal FTD can be accessed in different ways, each of them covering slightly different domains. The universe of Financial Transaction Data includes electronic and non-electronic transactions. Among electronic transactions we may distinguish from the following main types of FTD: MBWAY and MULTIBANCO (electronic payment and money transfer processes managed by SIBS, a payment services provider); Electronic transactions managed by Credit Card companies); peer–to-peer electronic transfers between banks; and data from the Tax Authority.

Data owners and data providers

The data owners consist of the private companies, and banks, and the Bank of Portugal. The access to data from private companies and banks is difficult since these companies are not obligated by law to report microdata to Statistics Portugal. The Bank of Portugal collects this information for the transactions data in aggregated way, and SIBS (who manages MBWAY and MULTIBNCO), manages almost all electronic transactions (over 60% or all transactions are made electronically and almost all are monitored by SIBS). The Tax Authority is an important data provider, as it collects the data for all the transactions in Portugal.

Process for getting FTD

The different transactions, B2B, B2C through bank cards, direct debits and transfers, payments, withdrawals and purchases are all monitored by the Bank of Portugal and the Tax Authority. After an agreement between Statistics Portugal and The Tax Authority, this will be the data provider for the project.

Promising statistics

Either as an additional source for improving the quality of statistics, or as a source that can replace the existing source we aim at focusing on turnover in specific sectors of the Portuguese sharing Economy, such as accommodation and transportation.

3.6 Slovenia

Types of FTD

- Central bank of Slovenia collects quarterly aggregated data on payment cards, cash withdrawals and deposits, electronic banking, credit transfers and direct debits.
- BANKART, private company owned by Slovenian banks, ensures processing of transactions with various banking payment instruments. As a result, they own data on ATM operations, card payments, payment systems (SEPA, SIMP-PS and E-invoice) and on POS terminals of majority of Slovenian banks.
- With fiscal cash register Financial Administration of the Republic of Slovenia monitors all issued invoices, which are paid in cash, credit and debit cards, and do not exceed value of EUR 5,000 for consumers (B2C) and EUR 420 for businesses (B2B).

Process for getting FTD

As Bank of Slovenia collects only aggregated quarterly data, this data is not relevant for Statistical office of Slovenia. We are in talks with both Financial Administration of Slovenia and BANKART regarding obtaining

the data on financial transactions. In time scope of this project, we are most likely to obtain data on fiscal cash registers.

Metadata

Obtaining data on fiscal cash registers would give us access to individual cash or card payments in terms of tax number and location of the issuer, time and date of issue of invoice and the value of the invoice, broken down by tax rates. Missing is the information on the payer, unless in case of B2B sale.

Promising statistics

The most promising statistics are statistics on retail trade. Currently we calculate indices on retail trade based on data from survey and administrative data on value added tax, which becomes available to us 45 days after reference period. As the data on cash registers would be available 10 days after reference period at the latest, we could get more accurate results at time of first publishing of provisional data, which is 30 days after reference period.

4. Recommendations

For an NSI to get financial transaction data from a potential data provider, a necessary part is to either have a decision based on the Statistical act, or if the Statistical act is not applicable, there must be a formal agreement between the NSI and a potential data provider. In the latter case, an institutional agreement with the National Bank typically is appropriate, since the National Bank often has access to more data than the NSI**.**

Under either of the solutions above, a prerequisite is that the data provider should in some sense be motivated for providing data to the NSI. Below we will consider some elements of motivation before we look at the different steps in a data acquisition process.

Good cooperation through a good relationship is essential for obtaining financial transaction data, whether the Statistical Act grants access to data or not. It is always more tempting for a potential data provider to use resources benefitting someone with whom you have a good relationship. The importance of a positive relation is increased due to the data provider's need to use resources in preparing data for the NSI, as well as sharing and explaining metadata. The NSI usually cannot pay for these expenses and should acknowledge the burden put on the data provider. The data provider should in addition get something in return, and the NSI should ensure telling about these benefits to the data provider. In addition to contributing to the development of official statistics in the society, the benefits could be to get reduced response burden from shorter or fewer official statistics questionnaires. In addition, the NSI can help improving metadata description, developing data infrastructure, or performing relevant analyses of the data. One approach to achieving the three latter benefits, is by letting NSI staff perform tasks within the data provider's office. The advantage is firstly that data security concerns can be more easily solved, and even access to microdata might be possible. Secondly, knowledge transfer to the data provider is a possible outcome.

A well-founded description of the society's need of the financial transaction data, is necessary for motivating the potential data provider to use resources on providing data, especially if the resources should be used within a rather limited calendar time interval.

The data provider is concerned about the secure handling of the data that they possess, especially since financial transaction data are widely considered as sensitive. Reassuring the data provider of the secure handling of data within the NSI is an essential part of the contract with the data provider. A related topic is to inform the data provider of the role of the NSI. Contrary to most other governmental organisations, the NSI are not interested in the contents of the transactions for a single person or a single business, but to use data for statistical purposes only. The purpose of the NSI is to deliver aggregated numbers.

Central steps in the process of getting data are the study of metadata, making a delivery agreement, and in the situation where no microdata can be obtained by the NSI: specifying how data should be aggregated before sending them to the NSI. The thorough study of metadata is an important step that could reveal that presumably interesting data are less interesting for the NSI. Not only the available variables and their descriptions should be studied, but also what is the unit of the microdata, which part of the population is covered, and known quality issues.

The process of getting data usually takes a lot of calendar time as well as personnel resources within the NSI. A thorough study of metadata can be time-saving in cases where this study leads to termination of a data acquiring process that would only end in data with little usability within the NSI.

PART II - Sharing economy platforms (SEP)

5. Definition of Sharing economy and Sharing economy platforms (SEP)

Sharing economy is a phenomenon as old as human beings are. People always shared possessions, efforts, etc., whenever these assets were available from their regular use. However, before the internet boom, people tended not to share with strangers outside their local community. With the internet expansion, sharing with strangers turned possible and the number of participants in the sharing of a single asset, as well as the variety of goods and services subject to sharing, increase substantially and lead to the current sharing economy that we now are interested in capturing in official statistics. Below, we will first discuss a theoretical definition of sharing economy. Then we will discuss different operationalisations.

The papers on sharing economy show that there is no definition of sharing economy shared by all economists and business people (Codagnone & Martens 2016; p.8), which might seem strange in light if the ancient origin of the sharing economy. Also, there are other terms that are partly overlapping with sharing economy, such as collaborative economy, access economy and gig economy. Technology development will enlarge the sharing opportunity. Think about distributed energy systems. They slowly gain importance with renewables penetration in energy production. The subject of sharing is a renewable source of energy: some roofs are sunny, some are not; some places are windy, some are not, etc. Some places could produce a lot of green energy and could share it with other consumers. Sharing economy develops over time and our definition should accommodate emerging models of sharing

We will adopt a definition of sharing economy that is suitable for the use of SEP data or other sharing economy FTD in the production of official statistics, either for improving, quality assessment or replacing existing sources for official statistics. Our basis for the definition, is the literature on sharing economy and SEP. However, there is a vast amount of literature on this topic, and our ambition is not to cover all relevant literature, but instead to study enough literature to be able to make a suitable definition.

A *theoretical definition* describes what should be contained in sharing economy and SEP. However, the *operational definition* is what can be used in practice for official statistics production and is thus focus in this section. The relation between theoretical definitions and operational definitions are studied in Appendix 8.

5.1 Theoretical definition

Let us start with one definition of sharing economy: Frenken & Schor (2017) defined sharing economy as persons *granting each other temporary access to under-utilised physical assets ("idle capacity"), possibly for money.* Important fundamentals of sharing economy are temporary access, idle capacity and that the ownership of the asset is not changed.

When using a spare seat of a car ride or using a car/apartment during a week when the owner doesn't use it, means using an asset that would otherwise not be used. The notion of underutilised capacity is central when drawing a border line between sharing economy and economy on demand. When the house owner rent out while being away on vacation or business trip, or he/she has one or more spare room, the asset is underutilised and renting it is a sharing transaction. When the owner buys a second house and rent it round the year this is an accommodation business and no sharing is involved. The same reasoning could be applied to ride-sharing.

The definition by Frenken & Schor (2017) points to certain patterns that may or not be defined within the scope of sharing economy and related economies:

- whether temporary access that is paid by money is a part of sharing economy: required (R), allowed (A), or not allowed (N).
- whether businesses should be allowed (A) to be one of the two peers in the transaction or not (N). In the latter case, both parties must be persons.
- whether non-physical goods (services) are allowed (A) or not (N)

In addition, we add a fourth pattern:

- whether the existence of a platform mediating between the two peers is required (R) or allowed (A)

Above, *required (R)* means that without the presence of this property, the transaction is outside the definition.

If we represent the definition's relation to these four patterns in a vector, we can represent the definition above as (A,N,N,A), since Frenken & Schor (2017) argues against allowing services, which they argue are *on demand economy* or *gig economy,* and which lack the property of using an asset that would otherwise not been used.

We will adopt the exclusion of service from the definition of sharing economy: When a person is not using a service, the service provider can use his/her time differently. Rising capital through platform is a relatively new way to rise funds from many supporters and by-pass traditional sources of capital like banks, venture capitalists, and likes. The motives of participants in the fund rising platform (capital seeker and supporters) are a mixture of fund raising, hefty return, expanding awareness, being part of a community, support a cause, etc. (Gerber & Hui 2013). None of these has to do with sharing underutilised assets.

Commission (2016) considers that 'sharing economy' is often interchangeable with 'collaborative economy' and its definition of the latter considers "…activities are facilitated by collaborative platforms…" for temporary usage of goods but also services, that "… generally do not involve a change of ownership and can be carried out for profit or not-for-profit". Its definition can be represented by the vector (A,A,A,R), where the 'R' means that any "activity" that is not mediated by a platform, falls outside of the definition. Thus, when a person A hires out her apartment to person B for a week directly, this is not a part of the collaborative economy.

We notice that both definitions above focus on making available under-utilised assets on a temporary base with no change in ownership. However, contrary to the definition of sharing economy, the definition of collaborative economy

- allows businesses to be one of the parties in the transaction.
- Ssmetimes allows change of ownership?
- allows services in addition to physical goods.
- insists on the presence on a mediating platform.

Since we have a focus on the shared economy *platforms (*SEPs*)*, it is natural for us to restrict our attention to the part of the sharing economy being mediated by a SEP, just as in the definition of collaborative economy. However, in contrast to collaborative economy, sharing economy is about sharing, and thus we adopt the requirement suggested by Frenken & Schor (2017) that a business cannot be one of the parties. For the same reason of focusing on sharing, we also choose to restrict our attention to physical assets. Thus, the only difference from the definition of Frenken & Schor (2017), is that we insist on the presence of a SEP.

5.2 Operational definition suitable for official statistics production

We need an operational definition that, in theory, lets us measure the shared economy using either FTD or SEP data. We then start by adopting an operational definition having the following additional requirements:

- the transaction for the temporal access should involve payment in money. The sharing is then done *for profit* which is here defined as covering at least the costs, e.g. fuel consumption.
- Either the mediator or one of the two peers involved, should be resident in the European Economic Area.

Remark 1: We could also choose to allow businesses to be provider, since it is difficult to separate businesses from persons in SEP data and FTD data – remember that an individual can be a business in the sharing economy context. If we allow businesses in this role, we adopt an *extended* shared economy definition.

Remark 2: If we choose the extended operational definition above, we could perhaps also consider the 'collaborative economy' definition for the work on measuring sharing economy, instead of the more "pure" sharing economy definition of Section 5.1.

We now have two options for the operational definition, given the suggestions of the two bullet points above. One is to extend the operational definition and to include business and services inside the scope of SE, which corresponds to defining sharing economy as the bluish circle of Figure 5.1. The other is to stick to the narrow definition, introduced at the start of Section 5.1, and not allow for business and services to be part of SE, thus restricting sharing economy to the clear blue part of Figure 5.1.

If we extend the operational definition by allowing businesses as well as services, we extend the sharing economy boundary and include transactions that constitute economy on demand, or gig economy. Thus, quantitative indicators based on the extended definition will greatly overestimate sharing economy on job creation, income generation and income distribution across sharing economy participants compared to the narrow definition of sharing economy (clear blue part of Figure 5.1). We choose to adopt the approach of starting with the narrow definition and then include new components after carefully weighting cons and pros for extending the sharing economy boundaries.

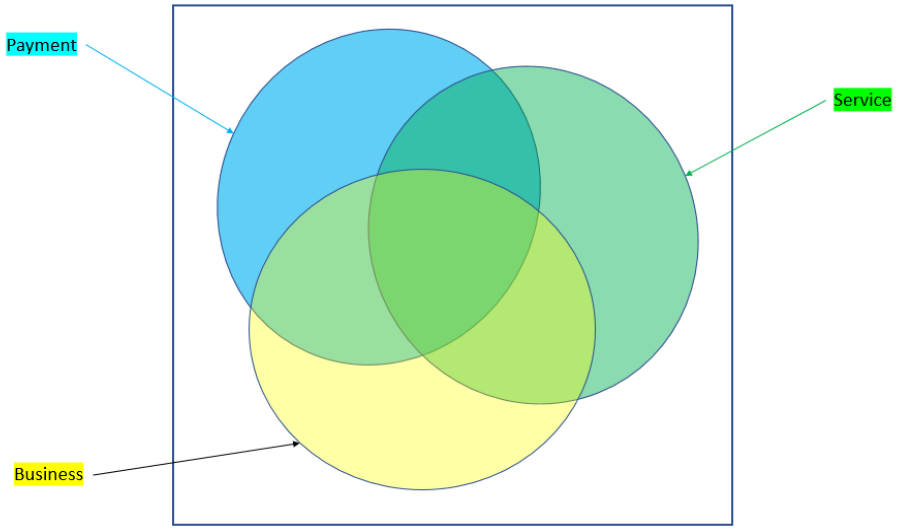


Figure 5.1. Illustration of the contents of the sharing economy concept depending its inclusion/ exclusion of payment (blue), businesses (yellow) and service (green).

5.3 More on definition of the platforms

Considering the scope of the term 'SEP' (sharing economy platforms), we will include any internet site that is acting as mediator of shared economy transactions. Notice, however, that from the perspective of the narrow definition above, many SEPs are also mediating in transactions not being part of the sharing economy, and some SEPs are even *mainly* intermediating in non-sharing economy transactions. Difference between Blablacar and Uber are instructive. Both companies provides shares ride services. However, business models are quite different. Blablacar employs cost sharing business model. Drivers makes no profit but share cost with strangers for gas and car wear and tear. Uber drivers are independent contractors who work for profit. Blablacar transactions fall into sharing economy domain, Uber transactions fall into on-demand economy domain.

6. Sharing economy platform (SEP) data: data accessibility and SEP-related indicators based on FTDs

6. 1 Introduction

By considering the difficulties in defining and delimiting the SEP domain for statistical purposes, the definition of a methodological framework aimed at identifying the potential sources for new SEP-related indicators (and designing an effective production process for them, from data access to data processing and quality checking) has to be seen as highly challenging.

Some questions might help to frame the issues currently at stake:

- Which platforms exist and if they can be classified into operational categories.
- Under which conditions SEP-related data can be accessed/used.
- Whether a set of SEP-related metadata can be produced.
- Which uses in official statistics are more promising for SEP-related data.
- Whether empirical analysis of one or more SEP-related FTD (Financial transactions data) datasets could be performed.

6.1.1 The options developed by the WPG members

Notice that in Section 5, the theoretical definition and more narrow operational definition of sharing economy on one hand, and the definition of collaborative economy on the other hand, represent two different points of view on what the SEP field is about:

- Activities whose main purpose is that of allowing the sharing of goods or under-utilised assets among individuals and households through the neutral (free) intermediation of technology platforms; or
- Activities performed, for profit, by technology platforms implementing a business model based on the intermediation, or sharing, between peers (a provider and a user, i.e. operating in a double-side market).

These two definitions imply addressing two distinct populations of actors in the SEP arena: a) non-for-profit, mostly locally based, web platforms dealing with C2C sharing activities for the former, b) large, for-profit, multinational (less commonly, national-only) platforms for the latter. Therefore, different strategies have to be taken into consideration in order to set up a consistent measurement framework:

- Limiting the analysis to 'pure' sharing platforms (possibly, at country level).
- Limiting the analysis to the largest digital platforms operating in a country, irrespective of where their headquarters are located.
- Combining the two populations.

A choice between these alternatives has great consequences in terms of methods and data sources to be used, as well as on the comparability of the results that could be achieved in EU countries (at both pilot and full-scale level). In principle, to bring the analysis to a too detailed level (e.g. a single region or a single sector within a country) will strongly affect the possibility to converge on a single model of quantitative measurement of the SEP phenomenon and will bring into the analysis some unavoidable biases. On the other hand, focusing on a small number of big players operating, for instance, in all (or almost all) EU countries could allow for the access to a common evidence base. The advantages of such input harmonisation in the

collection of platform data could be, of course, partially or totally balanced by the heterogeneity, often observed, in the use of the services delivered by international digital platforms across [8].

The availability of data should be the main criterion to be used for taking a decision about how to proceed, with a basic distinction between what could be defined a 'direct' and a related 'indirect' approach, c.f. Table 6.1 below. A 'direct approach' would imply directly addressing platforms with a request for data on their activities (at country level). The conditions needed to opt for this option will be discussed later.

Table 6.1 is aimed at describing how three factors affecting the measurement of digital platforms – namely, (a) the typology of platforms under observation; (b) the availability of direct vs. indirect indicators and (c) their technical feasibility - are influenced by different methods of data collection.

For instance, by comparing tow data collection methods: a survey on platforms' users and a collection of data from platforms, a number of differences could be observed. By definition, data collected form a given population of platforms allows for a full coverage of the target population, while a survey of users will be affected by biases according to the platforms actually used by respondents (less severe for a survey on local platforms, pretty relevant when targeting international platforms with user surveys at local or national level). Moreover, any data collection but the provision of data from the platforms themselves will produce "indirect indicators". Finally, it has to be pointed out that surveys are always constrained, in scale and scope, by the amount of resources available. On the other hand, to find an agreement with one or more digital platforms in order to share some data, more or less aggregated, about their business will imply a number of legal and managerial issues not easy to deal with.

Data potential and access to sources are key factors to be taken into consideration when defining a data collection strategy. Still with reference to Table 6.1, a preference should be given to sources that could provide relevant information as far as they are accessible to users.

Overall, it can be argued that large platforms make available a big amount of information on their activities that can be potentially addressed through a range of sources, while small local platforms need to be directly addressed to collect some relevant information. From a broader perspective, a trade-off has to be noticed between the potential of a direct access to platform data and the reluctance of platforms to allow such an access to users. This is an essential point to be considered when defining a data collection strategy: the more a direct access will become difficult, the more the alternative sources will become convenient and attractive in order to produce statistical indicators.

[8] With a higher comparability when data are centrally collected and checked by a single institution as currently tested by Eurostat with reference to the activities in EU countries of big multinational platforms offering accommodation and travel services.

Table 6.1. Target populations of digital platforms and potential data sources

	TARGET POPULATIONS				FEASIBILITY
DATA SOURCES	Small local platforms	Large multinational platforms	Both populations		
Surveys of individuals (users)	Partially useful	Limited use	Partially useful	Indirect indicators	Affected by resources and time constraints
Surveys of enterprises (providers)	Irrelevant	**Relevant**	Partially useful		
Public administrative data (e.g. fiscal data)	Partially useful	**Relevant**	Partially useful		Confidential issues to be considered
Financial trans. data (FTD)	Irrelevant	Partially useful	Partially useful		
Web data collected from platforms	**Relevant**	**Relevant**	**Relevant**	Direct indicators	Agreements hard to set
Web data collected by web scraping/mining	Partially useful	Partially useful	Partially useful	Indirect indicators	Data protection issue to be considered

This is the context where data sources can be compared in order to find the right balance between the resources needed to extract and process the data and the informative contents of the expected output.

Three options have been compared by WPG members in order to define a suitable strategy to produce SEP-relevant indicators in relation to FTDs:

a. Investigating if - in the literature - issues about SEP transactions are addressed with reference to: accessibility of data, data infrastructure and potential indicators.
b. Collecting SEP data from various data sources according to national interests and specificities.
c. Developing some FTD-based SEP indicators as a basis for a quantification of platforms' economic impact.

6.1.2 Option a. The literature review

This is the option initially envisaged when the WPG was approved. Lately, some questions were raised about the actual availability of a substantial amount of research activity whose results could be effectively used to meet the WPG requirements. By considering that all contributing countries will be involved in this activity, a mixed population of large and small (local) platforms could be probably targeted.

A further issue is that about how 'literature' could be used. In principle, only scientific literature could be helpful for deriving (non-systemic) indicators, as 'grey' literature or media sources do not match the basic requirements to be used as reliable sources for development of indicators in the official statistics framework[9].

[9] By taking AirBNB as an example, an assessment should be done whether commercial sources of AirBNB-related data, like AirDNA.co, or freely accessible repositories of (unauthorised) scraped data sources (like insideairbnb.com) could provide a useful basis for developing new official indicators by NSIs.

Thus, by considering both small and large platforms, a scientific literature review could provide useful - although very partial - material to support future NSIs activities about:

- Digital platforms' data accessibility.
- Metadata detailing the use of data for digital platforms' measurement.
- Development of indicators which could feed official statistics activities.

A preliminary screening of scientific articles made available in Science Direct over the last five years (2015-2019), carried out by Istat, gives a preliminary (and probably partial) picture of the most common options for producing quantitative evidence on the digital platforms' activities as a result of scientific research activities.

By focusing only on scientific articles describing issues related to the measurement of any phenomenon related to digital platforms (articles about jobs contracts, and jobs-related issues in general, have been excluded by considering them a field of its own), 48 articles have been identified. Among them, 35 articles are about AirBNB, 3 about Uber, 1 about the Mechanical Turk and 1 about Lyft. References to only-national or local platforms have not be found.

Not surprisingly, most of the articles are reporting about data accessibility, metadata and new indicators in a way that makes difficult to identify a specific focus on one of these three aspect of data production and analysis. By considering AirBNB-related studies, almost half of them originate from concerns about the economic and social impact of short-term house renting in large cities. The source of data mostly used is the legal scraping the AirBNB website. It is remarkable that other sources are used, as well: from surveys of AirBNB users, to social media where opinions on AirBNB are given and shared among users.

Although based on a limited evidence, the Istat preliminary review allows for arguing that a literature review on digital platforms could hardly provide a knowledge base that would be useful for developing reliable experimental statistics in the ESS framework. Some points can be highlighted.

- A systematic review of potential sources of data on digital platforms can be hardly based on a literature analysis because of the constrains researchers have in accessing relevant databases. By definition, a whole range of data sources – chiefly, administrative data – are only accessible by NSIs.
- Most of the research activity focuses on the 'local impact' of 'multi-national platforms'. The poor attention paid to local platforms is a minor issue compared to the limitation of such analyses to highly localised phenomena. This prevents a comprehensive approach to the measurement of platform-related phenomena which are, of course, locally differentiated (at least, as far as their impact is concerned) but that, nevertheless, have common features which should be addressed when defining a broad (European) statistical framework.
- For a methodologically sound exercise, the scientific literature in the field of economics (or, broadly, of social sciences) is of poor use as issues about how to process the available data are much more relevant in the literature than those about the extraction/collection of data from the original sources and their quality monitoring. Of course, data quality is an essential requirement when exploring new and unstructured sources of data for statistical purposes.

6.1.3 Option b. A non-systematic collection of platforms' indicators

In addition to a set of studies produced by the European Commission[10], some European NSIs – while awaiting for a harmonised initiative by Eurostat in this area – have started to address the issue of measuring the platform economy at national level.

[10] Several DG JRC papers can be mentioned (e.g.: Codagnone, C., Biagi, F. and Abadie, F. (2016). *The Passions and the Interests: Unpacking the 'Sharing Economy*. JRC Science for Policy Report, Institute for Prospective Technological Studies, EUR 27914 EN, doi:10.2791/474555; Codagnone, C. and Martens, B. (2016). *Scoping the Sharing Economy:*

In addition to a number of articles and press releases from EU's NSIs, two papers describe the state of the art of such efforts with reference to the UK and Dutch cases:

- Office for National Statistics (ONS) (2017). *The feasibility of measuring the sharing economy: November 2017 progress update*, London.
- Heerschap, N., N. Pouw & C. Atmé. (2018). *Measuring online platforms*. CBS in co-operation with UvA, Amsterdam.

Just considering these two papers, most of the methodological information that should have been extracted from a literature review is made available to users in a comprehensive and usable form. The logical process undertaken by the authors has been, in fact, that of reviewing all available sources - either under direct control by or made accessible to the NSI – of information on digital platforms (or any other activity related to sharing economy phenomena).

The evidence which can be found in these papers is more or less that available from most EU NSIs even though without any specific harmonisation effort. The implementation of a specific effort of methodological harmonisation at EU scale could substantially improve the prospects for these activities, mainly in specific areas of statistical production like tourism statistics.

In this perspective, the current activities by Eurostat of starting a process of co-ordination – restricted, at the moment, to tourism statistics – raise the question whether NSIs should invest in systematically collecting data on the activities of big platforms in their countries. In fact, these activities could be soon displaced as a result of a centralised data collection of big platforms' activity data under the Eurostat responsibility (already planned for and under negotiation with some digital platforms with activities all over Europe).

By excluding the methodological assessment of current NSI's activities, on the one hand, and the development of new methods of harmonised data collection with reference to big platforms, on the other hand, the purpose of the WPG activity could be refocused on assessing the availability of data on small, national or local, digital platforms. The argument for pursuing this approach is that this area (which should include the largest part of the so-called sharing economy) has been largely neglected although its relevant social, if not economic, impact.

6.1.4 Option c. Developing FTD-based SEP indicators

A third option taken into consideration to fulfil the requirements of WPG was that of assessing to what extent the two parallel streams of activity included in WPG could be bridged in some way. Two factors can be pointed out to support the idea of investing in this activity. First, the potential for exploiting synergies within the WPG group. As FTD experts and SEP experts are working shoulder to shoulder, the exchange of information and, even more, a common development of methods and practices is easier than with reference to any alternative source. Second, FTD have a potential as a source of indicators on SEP as financial transactions' management – in a context of large heterogeneity among platforms, peers and goods/services – is the only comparable process to be consistently undertaken by all platforms.

It has to be anticipated that the strict confidentiality rules to be enforced when processing FTDs will make almost impossible to extract a set of transactions (micro-data) based on the ID/bank account codes of single platforms. This implies that the targeting of a specific sub-population of firms managing platforms is barely possible. But large platforms can be hardly targeted by NSIs not just because strict confidentiality rules but

Origins, Definitions, Impact and Regulatory Issues, Institute for Prospective Technological Studies, Digital Economy Working Paper, 2016/01. JRC100369) together with the 2018 EC Report on *Study to monitor the economic development of the collaborative economy at sector level in the 28 EU Member States* (doi 10.2873/83555).

rather because of their multinational structure, almost impossible to monitor from a purely national perspective.

Some positive alternatives will include the targeting of 'platform users', 'platform providers' and transactions. The FTD-related statistical activities are based on the implementation of ad-hoc selection criteria which could allow for extracting the key information available from the FTD without breaching confidentiality. On the platform side, the proposal could be that of developing a process of aggregation for FTD which could make the aggregated information relevant to the platform economy measurement (i.e. its volume of activities and related GDP impact).

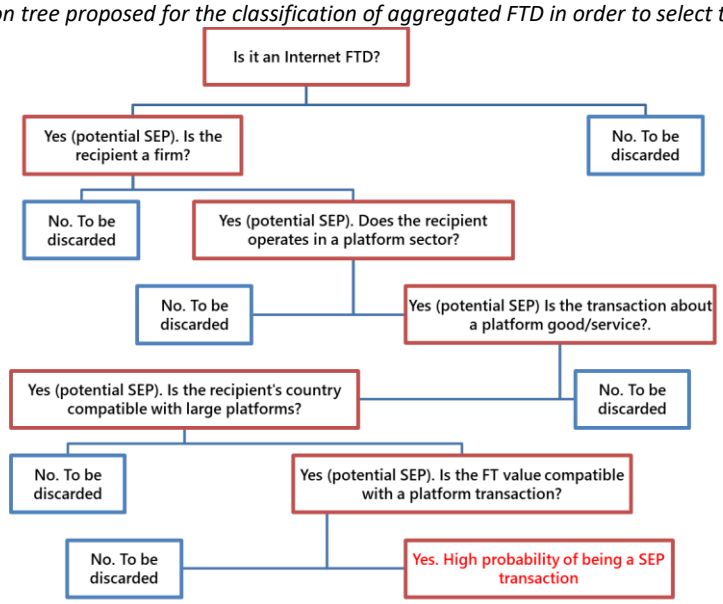
In order to make a methodological work aimed at linking FTD and SEP in line with the WPG work programme, the themes of accessibility, metadata and new indicators have to be properly addressed.

As far as the issue of the accessibility of SEP-related FTD data is concerned, some activities have to be undertaken, even though only at experimental level:

- focusing on Internet transactions (included e-commerce) and collecting available (aggregated) data with SEP potential, as a first step;
- highlighting specific patterns of online FTD, included sectorialisation, average value and international flows, as a second step;
- exploring to what extent online FTD could be more or less accessible, at micro-level, than other FTD, as a third step;
- testing the collection of related information on the use electronic payments online (Web analytics, e-commerce statistics, etc.), as a final step.

The focus is, of course, on aggregated data and on the availability of a detailed breakdown of FTD when coming to features which will be needed to identified SEP-related transactions. Similar accessibility tests should be carried out in all countries involved in the ESSNet (even beyond WPG contributors) providing they had achieved and even limited access to FTD and could be able to produce a list of variables available (with an associated level of breakdown) to describe FTD in the national territory. This comparative assessment will have to be based on a common methodology which could rely on a classification tree to make the selection easier and consistent across countries (Figure 6.1). The aim is that of identifying, as a by-product of the FTD work, those financial transactions which could involve – with a given probability – a large digital platform.

Figure 6.1 Classification tree proposed for the classification of aggregated FTD in order to select those FTD with a SEP potential.



As a second stage of the process, some specific methodologies have to be developed in order to deal with the SEP-related FTD data previously identified.

Classification have to be the first step. As an essential part of the methodology that will be developed in order to extract from the FTD population those observations with a high probability to be SEP-related, a number of concepts and related classifications have to be produced, tested and standardised.

New classifications have to address nature, sector and features of platforms, peers and transactions themselves (as implemented in the proposed classification tree).

Table 6.2. Main classification criteria of FTD to be codified in order to be adopted in the SEP FTD classification tree

Electronic / non electronic
Nature of peers (firms / individuals)
Sector of activity of peers
Good / services transferred
Nationality of peers
Value of transaction

A high level of co-operation is needed among the countries contributing to the ESSNet in order to assure that the classification framework will be realistic with reference both to the actual availability of FTD data in all concerned countries and with the same activities of digital platforms as they emerge from available indicators.

6.2 The operationalisation of "option c".

After having found an agreement among the WPG members about the feasibility of the above-described option c, a range of activities have been undertaken in order to develop a methodological framework to test the option c basic concepts.

This option is about measuring the SEP activities in terms of (monetary) transactions. It implies that only platform-mediated transactions with an economic value and a related transfer of money can be taken into consideration. In addition to this requirement, by adopting a definition of "sharing platform" which focuses on "digital platforms" (operating over the Internet) the connected definition of transactions will have to be restricted to "digital financial transactions", i.e. those taking place between Internet operators (including a digital platform as a mediator between a user and a provider both operating on the Internet).

The role of platforms in the e-commerce framework

This definitional framework raises the issue of the relationship between e-commerce – as a broad phenomenon – and SEP activities. By assuming that the e-commerce encompasses all economic transactions taking place electronically, a subset of them should include Internet transactions and, among them, a further subset of digital transactions mediated by platforms. This is relevant to the extent Internet financial transactions (IFTs) should be used to measure the platforms' activities. That is because IFTs are actually measuring, at a broad level, the rate of digitalisation of a (country's) payments system and, more narrowly, the diffusion of e-commerce in a given economy (as far as the financial transfers are related to commercial transactions).

Table 6.3. A simplified outline of the «Strategic assessment of the three pillars of e-commerce» in FOSTEC & Company GmbH (2021, Figure 4)

	Types of Actors		
	Online marketplaces	Third party e-retailers	Direct sales including affiliates
Examples of actors	Amazon ebay Alibaba	wayfair vente-privee home24.de	A brand-name eShop with different affiliates.
Key components	International ecosystem infrastructure Large customer base	Companies acting online only, with websites having a focused product relevance. Highly relevant target group	Online brand offering seamless purchasing options over different media channels. brand-strengthening affiliates

By assuming that could be possible to identify those IFTs taking place as payments for the provision of goods or services, the actors involved in the transaction processes are those included in Table 6.3. In the e-commerce framework described in Table 6.3, platforms can be identified both as “online marketplaces” and “e-retailers”. The former are big commercial platforms, mostly with a multinational scope, which are acting as intermediaries between multiple sellers and a very large population of potential customers. A key feature of them is that of controlling the flow of payments for the goods purchased through the platform. Thus, they can be associated to a transactions’ flow from a crowd to a single player (and, at a further stage, from the single player to many providers). In the category of e-retailers most of the other digital platforms acting as intermediaries in the commercialisation of goods and services can be found. The financial transactions associated to them are very similar to those already described for the big platforms but they differ from them as to the number of potential customers: no longer an undifferentiated crowd but specific groups of customers. The transactions to be associated with this group of actors will be linking, in terms of transactions, from “many to a few”.

Finally, some basic e-commerce activities can be identified by considering a third group of actors: those involved in direct sales through their own e-commerce websites. Under this case, platforms are not involved in the transactions which can be defined as relationships from “many to many”. As a consequence, when deriving some platform indicators from FTDs data, these transactions should be excluded as far as possible.

Table 6.4 The three patterns of e-commerce transactions' relationships and related indicators.

E-commerce actors	Transactions' relationship	Potential indicators	Conditions/constraints
E-commerce total	Any	Volume of e-commerce sales	Ability to target Internet (cards) payments. Identification of the nature of transfer by using MCC codes.
Online marketplaces	Huge number of customers transferring money to a few large online retailers (crowd to a few)	Volume of e-commerce sales intermediated by marketplaces	Identification and selection of bank accounts ID used by marketplaces to receive payments (by number of transactions).
		by destination country	If breakdown by country available, "crowd to a few" transfers to selected low-taxation countries could be targeted.
Third Party E-retailers	Selected groups of customers transferring money to a few specialised online retailers (many to a few)	Volume of e-commerce sales intermediated by specialised e-retailers	Combination of intensity indicators (number of transfers to a single bank account) and MCC codes.
Direct sales	Groups of customers transferring money to a large number of online sellers (many to many)	Volume of e-commerce sales by producers	Identification of C-to-B bank transfers. Best suited to target intra-national transfers. Low reliability.

The above arguing is condensed in Table 6.4 where the potential for developing e-commerce and platform indicators by using FTDs is compared. Under very demanding constraints, it can be assumed that FTDs have indeed a potential to provide some broad information about the commercial transactions generating online bank transfers, usually by card. Of course, the main condition is that of having access to a significant amount of FTDs (either at micro or aggregated level) and, more specifically, to data about online bank transfers by card. Additional information about selected bank accounts' IDs, MCC codes and cross-border transfer could help to finalise the develop of a set of statistical indicators. The potential availability of such data has been tested with reference to the six EU countries contributing to the WPG.

6.2.1 The WPG mini-survey on the availability of FTDs to develop SEP indicators

In the period September-October 2019, the national teams contributing to the WPG of the ESSNet Big Data have been asked to report in a comparable and detailed level about the actual (and expected) availability of FTDs which could be used to produce some indicators to be used as proxies to measure the activities of digital/sharing platforms in EU countries.

The mini-questionnaire included three sections, respectively about data availability and sources (1), data breakdowns (2) and complementary information (3). The blank questionnaire is attached as Annex 1.

Two countries have not answered the survey: Portugal and Slovenia. Portugal reported that the NSI does not have currently access to credit card data (thus, no information on online payments by card). While for other WPG countries the main sources of data are public or private financial institutions, Portugal is currently collecting, for the WPG purposes – directly from the Portuguese Tax Authority - a wide-ranging evidence including all the financial transactions related to all purchases taking place in Portugal regardless of the payment method used. It can be assumed that all types of transaction will be included: electronic payments

(ATM, bank transfers or Credit Card), as well as non-electronic payments (in cash) but without a detailed breakdown by type. Further investigation will be performed about the possibility to get in the future from the Portuguese Tax Authority a detailed information about the payment method of each transaction. For a similar lack of detailed information about online payments, as well as transactions by credit card, also Slovenia declined the request to join the mini-survey.

As a result, four countries were able to provide, at least partially, the requested information: Bulgaria, Germany, Italy and Norway (the questionnaire sent by these countries are attached as Annex 2, the results are compared in Table 6.5). It has to be pointed out that information on online transactions by credit card is available only in three countries (Bulgaria, Italy and Norway): Germany does not have payments cards transactions including detailed information as for example the Merchant Category Code (MCC), which limits the scope for empirical analyses on estimating shared economy transactions.
As mentioned, the first section of the questionnaire asked the WPG countries about data availability. Questions 1.1 to 1.3 were indeed about data sources. All countries show a preference to deal with a single data source even when the coverage of 100% of the requested information from a single data source cannot be assured.

FTDs are will be in most cases available at national level (i.e. for the whole country), at least with an annual frequency. Only Bulgaria, among the respondents, will collect just data on the aggregated 'value' of financial transactions; the other countries will access both data on value and number of transactions.

As to the availability of FTDs detailed data on online credit card transactions (a key condition to derive e-commerce/platform indicators from FTDs), only Bulgaria, Italy and Norway positively replied, thus answering also the specific questions in Section 2 about which information will available about credit card data.

On the 'payer' side of the credit card online transactions (question 2.1.1), it is very difficult to identify a breakdown achievable in all three concerned countries. Nature (individual/business) and nationality of the payers are inconsistently available between the respondents. Quite encouraging is the possibility to identify the country where the credit card of the payer has been issued in most cases.

Much broader is the coverage of the 'receiver' side (question 2.1.2). First, information allowing for a clear distinction between residents and foreigners will be given by providers in all three countries. Under some constraints, also the nationality of the receiver could be guessed. About discriminating individuals from enterprises, this seems to be very difficult for receivers, too. Nevertheless, as far as MCC codes will be made available (probably, not in Bulgaria) some additional information could be extracted from them.

By focusing on the 'transactions' (question 2.1.3), the availability of the information on their total value can be given for granted. About the nature of the transactions, the MCC codes should be the main source to exploit (hard to gauge, at this point, how much information could be extracted from them).

Moving to the question 2.2 of the questionnaire, the possible combination of different breakdowns is taken into consideration. In Norway there is a huge potential for matching payers' and receivers' data with reference to the transactions' value. This is also possible in Italy but only considering the 'nationality' of peers, not their nature (individuals or businesses). Unlike Italy, data matching in Bulgaria is only possible with reference to the peers' nature. A matching based on the peers' nationality is either impossible or uncertain.

A similar matching between peers but with reference to the 'domain' (commercial sector) of the transaction seems even more problematic. Actually, it is highly feasible in Norway but totally impossible in Bulgaria and severely limited in Italy by the limited availability of information on the nature of peers (much better it is the availability of data split by residents/foreigners).

When asking the respondents to make proposals for developing new indicators, only Norway was able to develop its own vision about what could be achievable, by proposing four indicators:

- Payments with Norwegian cards over the Internet in Norway and abroad, and payments by foreigners' cards over the Internet in Norway (coverage 100%)
- Country where the card has been issued, and country where the card has been used (coverage 100%)
- Merchant Category Codes (coverage 100%)
- Use of cards issued abroad on user locations in Norway broken down geographically by country, and for Norway also by postal code (coverage 100%)

Even though Italy was not in the position to develop any specific proposal for new indicators, it can be argued that – on the basis of the answers to previous questions – most of the indicators proposed by Norway could be also produced for Italy, at least on an experimental basis.

Finally, Bulgaria forwarded a single proposal that has to be mostly seen as a potential denominator for further indicators:

- Aggregated online payments across residents and across residents and non-residents (coverage 100%)

The last section of the questionnaire (Section 3) was dealing with a number of issues related to the expected delivery of new SEP-related FTD indicators. Overall, Norway is quite open to experimenting new approaches (also because of their actual experience in processing FTDs), while Italy and Bulgaria (and Germany) which are still struggling in the process of defining the contents of the datasets they will be able to access, are understandably more cautious.

For instance, all countries but Norway are pessimistic about using indirect information in order to identify online transactions (for instance, by restricting the analysis to online operators. Norway is the only country considering the identification of payments from mobile devices possible. Similarly, only Norway is open to an experimental identification of the 'top receivers', for instance by looking at the rate of concentration by MCC code (even with a geographical breakdown).

Two countries, Italy and Norway, are indeed optimistic about the future use of MCC codes to deduct the nature of receivers (much less that of payers). The remaining countries are either pessimistic or have not access to the MCC codes. Similar arguing affect the options to produce a data breakdown by the 'receiver's economic activity' and the 'transaction's domain' both strongly dependent on the availability and reliability of MCC codes.

Also the identification of the peers' nationality will be largely based on data availability (still under discussion in all countries). Whether extending the basic breakdown between residents and foreigners (the only option for Germany) will depend on both the amount of information which will be made available to NSIs and even on the number of observations available for each country to be taken into consideration as a small number of observation will make the information useless because of the confidentiality constraints.

Final topics considered in the questionnaire's Section 3 refer to:

- Expected frequency of data release: arguably yearly in order to get comparable results.
- Potential use of FTDs microdata: excluded by all countries but Norway.
- Timing (trends, seasonality, etc.) of FTD transactions: to be taken into consideration by Norway and Italy mostly relying on a potential daily delivery of transactions data.
- Last but not least, the quality issue – quite sensible when dealing with big data – has been apparently already addressed by all countries compiling new indicators based on FTD

6.3 Perspectives for future developments

As the final aim of this exercise is that of proposing a set of new indicators to be designed and tested as a by-product of the mainstream uses of FTDs for statistics, the results of the mini-survey have to be seen in a long-term perspective, mostly suggesting the use of FTDs as just a complementary source for the development of SEP indicators.

Some ideas can be developed around the potential of a subset of FTDs – those about online credit card transactions – to tell us how the e-commerce – and, partially, the narrower platform economy – is evolving in EU countries.

A general remark has to be made about the constraints of the ESSNet exercise. The NSIs contributing to the WPG are experiencing many difficulties to get access to the FTDs with a level of detail that could allow for the standard production of SEP-related indicators. Has clearly emerged from the mini-survey that fiscal administrative data or aggregated data on national or international payments are useless when considering the e-commerce phenomenon. On the other hand, a direct access to data stored by banks and other financial institutions issuing credit cards are essential to describe the underlying commercial flows.

Potential, new SEP-related indicators, either FTD-based or as a combination of FTD and other data, could be expected to share some common features (mostly shortcomings) like: to be highly aggregated; featuring a high error coefficient; being only partially comparable across sectors and countries. On the other hand, they could have, nevertheless, the potential to be very timely and highly comparable over time.

As to the comparability across countries is concerned, the mini-survey results suggest negative prospects for it, at least without a comprehensive framework which could allow a number of countries to access the same, or highly comparable, data source. From Table 6.5 can be drawn the message that, so far, only indicators with comparable data for two countries can be proposed (missing the target to have at least three countries able to produce a small number of harmonised indicators). Some examples are given in Table 6.5.

Table 6.5. Proposed SEP-related FTD indicators for international comparison (only indicators available in at least two countries are included)

Indicators	Comments	Norway	Bulgaria	Italy
Aggregated total **value** of FTD credit card data transactions on an annual basis	Available in all countries			
Aggregated total **number** of FTD credit card data transactions on an annual basis	Available also in Germany	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis	Proxy of the total volume of e-commerce in a country	Yes	Yes	Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis	To be used as a denominator of more complex indicators	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by resident vs. foreigner **payers**	Possibly relevant for the purchase of tourist services in the country by foreigners		Yes	Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by nationality of **payers**	As above (with a breakdown probably limited to a few large countries)	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by nationality of **payers**	With a breakdown probably limited to a few large countries	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by resident vs. foreigner **receivers**	Key indicators to assess the relevance of foreign service providers and digital platforms	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by resident vs. foreigner **receivers**	Same as above	Yes		Yes

Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by nationality of **receivers**	Essential to target platforms based in selected countries (with a breakdown probably limited to a few large countries)	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by nationality of **receivers**	Same as above (with a breakdown probably limited to a few large countries)	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **economic activity** of **receivers**	Economic activity breakdown only based on MCC codes. Potential matching with economic statistics	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **economic activity** of **receivers**	Economic activity breakdown only based on MCC codes. Potential matching with economic statistics	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **type of purchase**	Type of purchase breakdown (e.g. goods/services) only based on MCC codes. Potential matching with economic statistics	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **type of purchase**	Type of purchase breakdown (e.g. goods/services) only based on MCC codes. Potential matching with economic statistics	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' nature** and **receivers' residence**	Key indicator for e-commerce (B2B vs. C2B). Partly related to platform activity	Yes	Yes	
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nature**	Key indicator for e-commerce (B2B vs. C2B). Partly related to platform activity	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nationality**	Mostly on cross-border purchases online	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' economic activity**	Key indicator for e-commerce	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nature** and by **type of purchase**	Key indicator for e-commerce	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nature** and by **type of purchase**	Key indicator for e-commerce	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nationality** and by **type of purchase**	Key indicator for e-commerce and platform activity	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' nationality** and by **type of purchase**	Key indicator for e-commerce and platform activity	Yes		Yes
Aggregated total **value** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' economic activity** and by **type of purchase**	Key indicator for e-commerce	Yes		Yes
Aggregated total **number** of FTD credit card data transactions **online** on an annual basis by **payers' residence** and **receivers' economic activity** and by **type of purchase**	Key indicator for e-commerce	Yes		Yes

The indicators listed in Table 6.5 could be used as a reference to develop a measurement framework to be used to quantify e-commerce and SEP-related activity in a country. Of course, the process is definitely an iterative one as every step in the combination of different pieces of information (generating indicators) shed

some light on specific features of the e-commerce and SEP phenomena and makes available additional knowledge to improve the understanding of the other indicators. It has to be stressed that this exercise cannot be considered valuable per se but as a component of a broader framework for the statistical measurement of Internet-related phenomena.

On the other hand, it is worth considering that the official statistics indicators on the e-commerce and the platform economy already available are still under development and that a substantial improvement of statistical methodology will be needed before achieving a sufficient level of harmonisation between economic and social indicators about SEP. SEP FTD indicators, in this respect, can be expected to face with the same issues currently affecting all other SEP-related indicators.

Finally, when considering the indicators described in Table 6.5 with reference to the activities of digital platforms (or SEP) some observations have to be done about what is actually measurable by using FTDs. Some evidences are, indeed, already clear:

- Total value of SEP-related transactions can be drawn by considering either (option 1) the single receiver with a high number of incoming online payments from individuals (less from businesses), or (option 2) by considering the online payments from individuals to a small group of countries where the largest digital platforms are based.
- To the extent the MCC codes will be able to allow for an accurate classification of transactions by type of purchase and by economic activity of the receiver, the SEP sector could be identified and measured, as well. It has to be highlighted that FTDs can only give information on the for-profit platform activities, thus neglecting any activity related to pure sharing purposes.
- A close monitoring of international payments, at an adequate level of granularity, could allow for the identification of SEP-related money flows, currently seldom targeted but essential for a range of purposes from taxation to statistical measurement. Unfortunately, NSIs can play a role in the process only if these data will be previously collected by other public institutions for administrative uses.
- Because of the previous point, an impact analysis of SEP-related domestic turnover (i.e. the share of retail activity controlled by digital platforms) will be possible only with very detailed FTD which are not expected to be made available to NSIs in the short term.

A final point to be addressed is the operationalisation of the indicators selected through the comparison of available data across countries (Table 6.5). This effort should be included among the pilot studies carried out in the WPG framework, possibly by sharing the responsibility between the three countries with the comparatively higher potential in this area: Bulgaria, Italy and Norway.

As soon as FTD quantitative evidence should be made available to the concerned NSIs, some experimental figures should be shared with the other WPG members in a process of convergence which should lead to the production of a short report including both a section with the available indicators and a section with a description of the methodology used as well as any issue related to the international comparability of the available indicators. Additional topics to be covered will include the options for extending the exercise to other EU countries and for producing consistent time series.

7. Conclusions

An inventory of the FTD infrastructure has been performed for the different countries of WPG: Bulgaria, Germany, Italy, Norway, Portugal and Slovenia. Also, the possibility of extracting information on sharing economy from the FTD, has been investigated.

We have seen that the infrastructure of FTD is different in different countries although there are some similarities. For Portugal and Slovenia, the only source that can provide access to FTD is the tax authorities. For Italy and Norway, banks, card companies and service providers are potential sources, but for Italy, this access must go through the National bank who by legislation has the right to access microdata and can produce aggregated data for Istat.

Only in Norway and Portugal is micro-FTD going to be available for the NSI. The reason for this is the differences in the legislation. It is worth noticing that micro-FTD from the complete population are large data and the NSI must have enough server capacity, and also a good system for pseudonymisation of any identificatory variables. Further, FTD are widely regarded as sensitive, and thus a GDPR requirement is that the NSI makes risk assessments before getting such data. One element in such assessments is to consider the need for official statistics against the perceived burden of the persons whose transactions are included in the FTD.

The need for most countries to only access aggregated data, means that an important process lies in getting the provider to aggregate the data in a way being useful for the NSI. This is a part of the cooperation that is needed with data providers to agree on data delivery to the NSI and comes in addition to having the necessary legislation for getting data. The data providers need to use resources on preparing data for the NSI, and the NSI in most of the countries cannot compensate for these resources. Thus, a good cooperation between the NSI and the data providers is essential for a successful outcome. Even in the presence of a good cooperation, the access to FTD takes some time.

Sharing economy is an increasing part of the economy. We have laid a framework for extracting information on sharing economy from the FTD using indicators on sharing economy activity. Further, we have investigated what relevant data exist in the countries for being input to such indicators.

This interim report will be updated in the final report of WPG in the autumn of 2020 where also several case studies will show empirically how official statistics can benefit from FTD.

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Appendices

Appendix 1. Terminology related to financial transactions

- *Acquirer* A bank or financial institution to which the acceptor (usually a merchant) transmits the information necessary in order to process card payments. When cash withdrawals take place over the counter or from ATM's, it is the entity (bank or financial institution) which makes cash available to the cardholder (either directly, or via the use of third-party providers).
- *Automated clearing house (ACH)* an electronic clearing system in which payment orders are exchanged among participants (primarily via electronic media) and handled by a data-processing centre. The ACH is pan-European (PE-ACH) in case of a platform and rules for the processing/clearing of euro payment instruments.
- *Automated teller machines (ATMs)* are terminals that allow authorised users, typically by using a card, to access a range of services such as cash withdrawals, balance enquiries, transfers of funds and/or acceptance of deposits. Note that not all ATMs need to have cash withdrawal functionality.
- *Cards* are payment instruments based on a unique number that can be used to initiate a payment, cash withdrawal or cash deposit that is processed using/over a card scheme or – for withdrawals and deposits at the ATM – within the network operated by the issuer of the card. The number can be stored on a plastic card (one of the traditional meanings of the word "card"), on another physical device (e.g. key tag, sticker, smartphone) or can be held virtually without a physical device.
- *Card schemes* are a single set of rules, practices, standards and/or implementation guidelines for the execution of card payments.
- *Cheques* are negotiable payment instruments based on written orders from one part (the drawer) to another (the drawee, normally a bank) requiring the drawee to pay a specific amount from a specified transactional account held in the drawer's name with that institution to be drawee, or a third part specified by the drawer. Cheques may be used for payments as well as settling debts and withdrawing money from banks. Cheques include traveller's cheques and banker's drafts.
- *Chips* are microprocessors embedded on a physical device, which are loaded with the information necessary to transmit payment information.
- *Contactless technology* allows for the transmission of payment information from a physical device to point pf sale (POS) terminals or ATMs without the need for physical contact between the physical device and the terminal. Information used for contactless payments can be stored on a chip, plastic card or other physical device (e.g. smartphone, key tag, sticker, watch). For the purpose of this data collection, the concept of contactless extends beyond that of near field communication (NFC) and also includes other forms of technology applied to perform contactless payments, e.g. Bluetooth low-energy and QR codes.
- *Credit transfers* are payment instruments based on payment orders or possibly sequences of payment orders made for the purpose of placing funds at the disposal of the payee. Both the payment orders and the funds move from the payer's institution to the payee's institution, possibly via several other institutions as intermediaries and/or one or more payment systems.
- *Direct debits* are payment instruments based on preauthorised debits, possibly recurrent, of the payer's account by the payee.
- *End users* are users of payment services as opposed to institutions offering payment services, for example consumers, corporates and merchants. The institutions providing payment services are also end users whenever they use payment services offered by others for their own retail payments (e.g. utility bills and salaries).

- *E-commerce* defined generally as the sale or purchase of goods or services, whether between businesses, households, individuals or private organizations, through electronic transactions conducted via the internet or other computer-mediated (online communication) networks. The term covers the ordering of goods and services that are sent over computer networks, but the payment and the ultimate delivery of the goods or service may be conducted either on- or off-line.
- *Issuers of payment instruments* are entities that (i) make payment instruments directly available to end users, (ii) authorise payments (on physical or online points of sale, over the counter or from ATMs) and (iii) in the case of card payments, guarantee payments to the acquirers.
- *Merchant category code (MCC)* is a four-digit number listed in ISO 18245 for retail financial services. MCC is used to classify the business by the type of goods or services it provides[11].
- *Non-banks offering payment services/instruments* are entities not licensed as banks according to their jurisdiction's legal framework which offer payment services, such as payment institutions, e-money institutions etc. This may also include the government if it offers payment services.
- *On-us payments* are payments in which the payer's and payee's accounts are held by the same institution. In the case of card payments, this means that the issuer of payment cards and the acquirer are the same institution.
- *Payment scheme* a single set of rules, practices, standards and/or implementation guidelines agreed between payment service providers for the execution of payment transactions across the Union and within Member States, and which is separated from any infrastructure or payment system that supports its operation.
- *Payment instruments* are instruments that give access to money, and are used for fund transfers, i.e. credit transfers, direct debits, card payments and cheques by use of bank account money, credits as well as e-money.
- *Payment services* refer to any service that is required for payments or for facilitating them.
- *Payment service provider: a bank, financial corporation, payment institution or e-money institution licensed for* offering payment service.
- *Payment system* this term has two meanings: 1) in some cases, it refers to the set of instruments, banking procedures and interbank funds transfer systems which facilitate the circulation of money in a country or currency area; 2) in most cases, it is used as a synonym for "funds transfer system".
- *Point of sale (POS) terminals* are devices typically used at a retail location to capture payment information electronically and – in some cases – on paper vouchers.
- *Prepaid cards* are a card on which a monetary value can be loaded in advance and stored either on the card itself or on a dedicated account on a computer. Those funds can then be used by the holder to make purchases.
- *Processing* the performance of all the actions required in accordance with the rules of a system for the handling of a transfer order from the point of acceptance by the system to the point of discharge from the system. Processing may include clearing, sorting, netting, matching and/or settlement.
- *Retail payments* are payments where at least one of the parties is an end user. This includes payments by institutions offering payment services when they use payment services offered by others to pay their own utility bills, salaries etc.
- *SEPA (Single Euro Payment Area)* process initiated by European banks and supported, inter alia, by the Eurosystem and the European Commission with a view to integrating retail payment systems and transforming the euro area into a true domestic market for the payment industry.

[11] https://www.iso.org/standard/33365.html

- *Settlement* is the completion of a transaction or of processing with the aim of discharging participants' obligations through the transfer of funds and/or securities. A settlement may be final or provisional.

Appendix 2. Bulgaria: financial transaction data (FTD)

The more monetised an economy is the more helpful FTD are in enriching existing statistics on domestic and foreign demand. Monetisation refers to the size of financial sector in the GDP. Economies with huge financial sector has a lot of financial instruments: cash, loans, tradable and non-tradable securities, etc. A deep financial sector provides a huge variety of financial instruments to business and households and less cash is used in transactions. The most obvious example is paying with plastics even small purchases like bus ticket.

Bulgarian is a rather poorly monetised economy and cash dominates in payments. According to Bulgarian National Bank POS payments and online payments for goods and services amounted to BGN 10.1 billion in 2017. (EUR 1 = BGN 1.956). The private consumption amounted to BGN 61 billion. In 2017. Assuming that all POS and online payments associate with consumption these payments still account for only 16 percent of overall consumption expenditures. Most likely several companies pay through POS terminals and/or online for goods and services supply (intermediate consumption): water supply, electricity supply, internet and mobile services, etc. It means that BGN 10.1 billion refers to both intermediate and final consumption. Hence, more than 84 percent of private consumption is paid in cash. A huge chunk of cash payments makes FTD not very informative about private consumption. Why? Because cash payments, unlike online and plastics payment, do not leave digital footprint in settlement system. Cash payments leave digital footprint in the national tax authority servers. In Bulgaria, each cashier's machine is connected with National Revenue Agency servers. When cash payments dominate the economy tax authority is a much better source of information than FTD.

A2.1 The different FTD

Available financial data

It is reasonable to consider the available financial data before moving to more detailed FTD. Bulgarian national bank publishes regularly data about monetary aggregates, credit to government and non-government sector (non-financial enterprises, households, and financial enterprises). The following monetary data are readily available:

- Monetary aggregates (M1, M2, M3)
  M1 consists of cash in circulation and time deposits. This is the most liquid part of money stock in an economy.
  M2 includes M1 plus demand deposits
  M3 includes M2 plus up to two years tradable debt securities.
- Loans to non-financial enterprises by amount and economic activity
- Loans to households and Not-for-profit-institutions-serving-households (NPISH) by amount
- Deposits of non-financial enterprises by amount and economic activity
- Deposits of households and NPISH by amount
- Card payment through POS terminal and online payments

Loans to and deposits of households and NPISH includes two institutional sectors: households and NPISH. Household sector consist of individuals/group of individuals as consumers. Also household sector includes households that produce goods/services for market or for their own final use.

Publishing data about households and NPISH is a shortage of banking statistics. It would be better to have data about bank savings of households only. Household sector is the net creditor in an economy and it is good to have data about savings and financial assets and liabilities of the sector excluding NPISH.

Deposits include overnight deposits, time deposits and deposits redeemable at a notice. Loans include overdraft; loans other than overdraft and repurchase agreement.

A2.2. Getting FTD

Until now, Bulgarian NSI has gained no access to FTD outside public data published regularly by National Bulgarian Bank. We get in touch with the Bulgarian National Bank and discussed possibilities to gain access to more granular data about C2B payments. No result until now.

Looking at the legislative framework it seems Bulgarian NSI should include the FDP based indicators in the Annual statistical program and passed it through the Parliament.

Art. 20 of the Bulgarian Statistical Act (cf. Section 2.2) addresses two important aspects of data access: first, databases we are interested in should be stipulated by a law and second, the survey that relies on that data bases should be included in the National Statistical Program.

The very first step toward access to FTD is to define indicators based on FTD, to put them in the National statistical program and to pass the program through the government. Having FTD included in the National Statistical Program increases chances to gain access to granular FTD.

A2.3 Metadata

The data about claims and liabilities of financial sector are reported by all banks in Bulgaria, including foreign banks branches. Reporting period is one month for monetary survey and quarter for deposits and loans by amount and economic activity.

Bank statistics is in line with the ECB requirements on monetary and financial statistics and with ESA 95. The economic activity breakdown follows Statistical Classification of Economic Activities in the European Community (NACE Rev.2).

Bulgarian National Bank collects and submits to the ECB statistics on POS payments and online payments on the annual base in line with the Regulation (EU) 1409/2013.

Data source of card payment statistics are licensed payment service providers on the territory of the country; the settlement finality payment system operators licensed by the BNB; the Real-time Interbank Gross Settlement System (RINGS) and the national system component of the Trans-European Automated Real-time Gross Settlement Express Transfer system (TARGET2-BNB).

A2.4 Official statistics potentially benefitting from FTD

The list of monetary data in Section A2.1 could prove helpful in improving quality of official statistics. For example, credit dynamic provides useful additional information to national accounts: GDP growth, investments, consumption, etc. In Bulgaria private credit to GDP ratio tends to decrease over last couple of years. This is an indication for poor financial intermediation and non-sustainable growth.

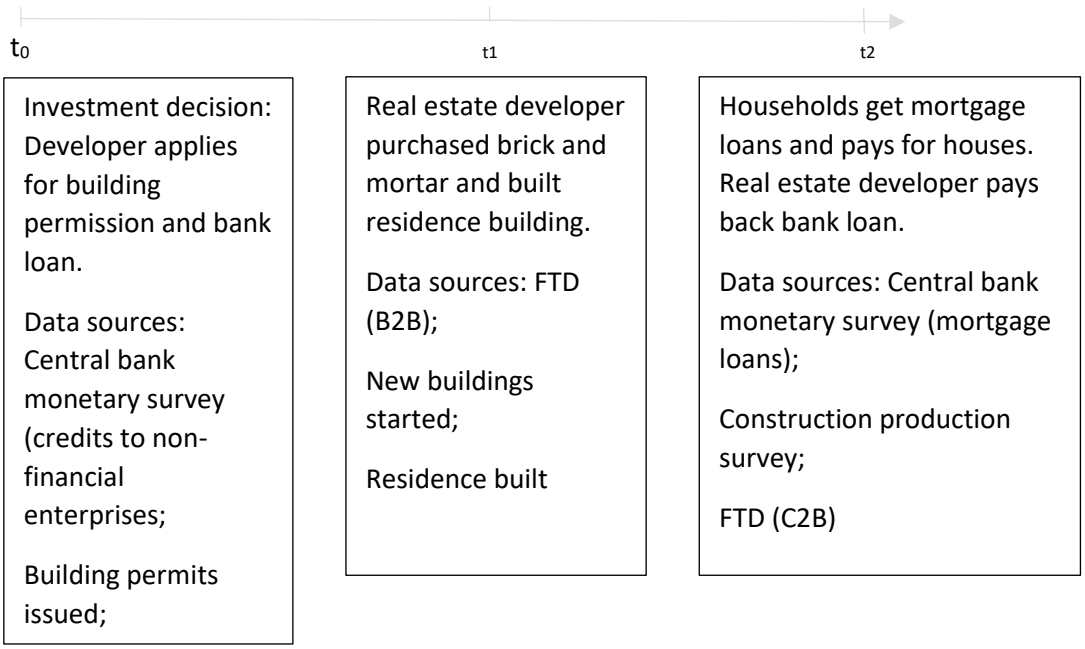
Deposits of households and NPISH provide useful information about saving rates and inequalities of households in terms of bank savings. In Bulgaria deposits and cash are the main instrument for saving because of the shallow financial market and small number of financial instruments. From deposit data available we see that 98 percent of deposits are up to BGN 50 000 (EUR 25 510). Those deposits accounts for 56 percent of the overall money in households & NPISH deposits. Only 0.01 percent of deposits are bigger than BGN 1 000 000. This small percentage accounts for 4.5 percent of overall deposits value. Quite unequal

distribution of deposits! Having in mind that many rich people save abroad the inequality is even bigger. One could easily park her/his savings abroad when domestic market is underdeveloped and do not provide a variety of instrument to save in.

The concept of inequality is multifaceted. People are unequal in terms of income, education, qualification, health, financial and non-financial assets possession, clean environment, etc. Deposits distribution is only a fraction of inequality phenomena. Understanding inequality implies a set of indicators rather than one indicator. FTD are informative only in the context of a broader set of indicators. Let us consider the following example.

A real estate developer decided to invest in a residence building. Before purchasing bricks and mortar, she/he applies for credit and after business plan approval the bank extends a loan to entrepreneur. After having the loan entrepreneur hires a construction company and pays for bricks, mortars, etc. Later on households get mortgage loans to pay for houses and developer pays back bank loan. Households repay mortgage loans within decades.

We can present the whole residence building process alongside time axis.



There are at least three players in the residence investment: real estate developers, construction company, and households. A data set could capture overall process from investment decision up to the buying houses by households in a more comprehensive way than FTD alone. Data about real estate developer activity address the supply side of the transaction or resources from the national accounts perspective. Data about households' mortgages and payments for residence refer to demand side of the real estate market. FTD are a part of a set of data that capture both supply and demand of residence buildings market. They could be very useful for National account department when aggregate primary data in Goods and services account, consumption and investments.

FTD have a good potential to show what is going on in the economy when considered within a broader set of statistics. Moreover, it is possible to detect a mismatch between supply and demand when looking at a set

of indicators. Suppose, there are too many building permits and loans extended to real estate developers. And there are not much mortgage loans. May be supply is above demand? Or this is a temporary phenomenon? The financial crisis in 2007 originated from huge mortgage loan extension and overheated real estate market. When supply greatly outstripped demand house prices collapsed, many homeowners found themselves under water, bad loans increased, and crisis erased quite a piece from banks, investors and households' balance sheet. May be a set of indicators could be helpful in detecting a mismatch between supply and demand on construction market and other markets.

Appendix 3. Germany: financial transaction data (FTD)

A3.1 What transaction data exist?

The *Deutsche Bundesbank* publishes annual time series on the number of charge and credit cards circulating in the economy and on the value of transaction. Since 2007, the number of those cards has increased by more than 60%, while the amount of transaction value paid with charge and credit cards has increased by more than 170%. The *Deutsche Bundesbank* publishes data about the number of card transactions and payments to POS terminals, while the Statistical data Warehouse by the *ECB* publishes data for the EU about payment card functions and accepting devices, and about payments per type of payment service. The available data are however only at the aggregate level and do not inform about the nature of purchase done per transaction or payment.

A3.2 Process for getting FTD

Like all euro area central banks, the *Deutsche Bundesbank* is part of the TARGET2 system. About half of the payments in terms of volume and over one-third in terms of value are submitted via the *Deutsche Bundesbank* through the Target2 payment system, with the German system called TARGET2-Bbk.

FTD relates to individual and retail payments and hence operate under TARGET2, under the TARGET Instant Payment Settlement (TIPS) and under the Retail Payment System (RPS). Under the TIPS, electronic retail payment solutions are processed almost immediately through interbank clearing of the transaction and the payee's account is credited. The settlement happens in secure central bank money.
For non-urgent payments, the *Deutsche Bundesbank* operates the Retail Payment System (RPS) platform. The RPS comprises of i) the cheque clearing service for the processing of euro-denominated cheque payments between credit institutions domiciled in Germany and ii) the SEPA-Clearer for the processing of euro-denominated national and cross-border transfers, direct debits and card payments.

Two Systemically Important Payment Systems (SIPS) are available through EBA Clearing: i) EURO1, a unique Real Time Gross Settlement (RTGS) equivalent net settlement system, held to the highest oversight requirements and overseen by the European Central Bank with the participation of National Central Banks of the Eurosystem and ii) STEP2, a central clearing house, provides banks across Europe the channel through which SEPA Credit Transfers and SEPA Direct Debits can be sent and received. The *Deutsche Bundesbank* participates in the Eurosystem's joint monitoring of the STEP2-T retail payment system operated by EBA CLEARING under the lead of the European Central Bank in its capacity as a competent authority. For instant payment activities, EBA clearing launched RT1, an infrastructure solution for the processing of instant SEPA credit transfers at pan-European level.

A3.3 Metadata for FTD

- Bundesbank data

In addition to the data passing through the payments platforms stated in the previous sections the *Deutsche Bundesbank* currently receives monthly data on credit and debit card turnover in relation to travel (balance of payments) pursuant to section 70 (1) 4a and 4b of the Foreign Trade and Payments Ordinance. This includes payments cards transactions of German residents travelling abroad as well as payments cards transactions of foreign travellers in Germany. However, more detailed information as for example the Merchant Category Code (MCC) is not known. The current data therefore does not allow for empirical analyses on estimating shared economy transactions.
However, the European Central Bank is currently revising the ECB regulation on payments statistics. Balance of Payments requirements may be imposed in the context of this update in particular with the intention of using payment card information to compile the travel item.

- Schufa data

The Federal Statistical Office also contacted the Schufa Holding AG, a German private credit bureau holding credit rating information about 67.7 million persons and 6 million companies for possible FTD data.
Schufa informed that they own some data, amongst others: the existence of a credit or debit card and its creation date, as well as the limit with revolving credit. However, their data does not inform about transaction data using credit and debit cards and can therefore not be considered for this exercise.

- Authorised payment service providers

A possible future data collection could emanate from authorised payment service providers. Those now fall within the second Payment Services Directive (PSD2) scope of applicability and are subject to supervision and monitoring by the National Federal Financial Supervisory Authority, the BaFin in Germany. A list of payment and electronic money institutions authorised or registered within the European Union (EU) and the European Economic Area countries (EEA) are available on the website of the European Banking Authority[12]. In future work it could be examined whether this may provide indications on representative payment service providers.

A3.4 Official statistics potentially benefitting from FTD

The collaboration between the Deutsche Bundesbank and the German Federal Statistics Office allowed evaluating potential future FTD statistics for official statistic production.

The type of statistics that would benefit from FTD depends on the level of detail, the population and coverage of FTD that would be available. Given that no German financial transaction data is currently available that would allow for empirical analyses on estimating shared economy transactions, it is not possible to give concrete examples of official statistics that could benefit from German Financial transaction data.

[12] https://eba.europa.eu/risk-analysis-and-data/register-of-payment-and-e-money-institutions-under-psd2

Appendix 4. Italy: financial transaction data (FTD)

A4.1 Types of FTD

The Bank of Italy directly manages the information on the interbank retail payment systems, through BI-COMP clearance and TARGET2 settlement systems. This information does not include some payment and settlement data directly managed by private entities (e.g., international credit and debit cards networks).

A4.1.1 FTD controlled by the Central bank

BI-COMP is a multilateral clearing system created in the late 1990s to guarantee the settlement in central bank money of euro-denominated retail interbank payments. The Bank of Italy provides the management of BI-COMP as a non-profit public service. BI-COMP handles the transactions and the bilateral balances between banks entered by the clearing systems and operates a multilateral clearing in six daily compensation cycles. At the end of each cycle, BI-COMP calculates a multilateral credit or debit balance for each participant that is sent for settlement in central bank money on the accounts held by intermediaries in TARGET2 (Trans-European Automated Real-Time Gross Settlement Express Transfer System). Preceding the multilateral clearing of electronic payments, there are activities (transmission, matching, confirmation of payments and computation of bilateral balances between banks) carried out by automated clearing house operating in a regime of open competition. Four private automatic clearing houses (SIA, NEXI and 'NEXI Instant and ICCREA') operate in Italy, whereas CABI is operated by Bank of Italy.

A4.1.2 FTD controlled by private operators

The EBA-clearance system operated by SIA and NEXI (the main card processors in Italy) is necessary in addition to the national central bank system in order to cover a sufficient part of the population through the clearance process. According to the last financial statements[13], SIA managed the processing of over 70 million cards per year, with over 1.1 million of points of sales; NEXI managed about 40 million payment cards and about 0.9 million points of sales.

The card processors are an alternative data source and their data benefits from the high data coverage (over 85% of the market) and the central management of the database (in terms of data reporting, aggregation, privacy issues, etc.). The internal archives of the (private) payment data processing entities have granular and timely payment data, which manage on behalf of banks. Data are collected on a daily (and also infra-daily) basis and contain information on the acquiring side (incoming payments by merchants) and the issuing side (outgoing payments by cardholders). On the issuing side information on the cardholder is not always available for the processor because it is usually only in the internal database of the originating banks.

However, data collection from card processors requires an intense experimental phase before proceeding with their exploitation for statistical production purposes, because problems related to either the harmonisation of different IT platforms or the set-up of ad-hoc data lakes might arise.

A4.2 Getting FTD

A4.2.1 The Bank of Italy - Istat collaboration agreement

The Italian Institute of Statistics has undertaken a collaboration agreement with the Bank of Italy for the exchange of aggregated statistical information. Istat has the possibility, through this agreement, to get information from electronic payment transactions available at the Bank of Italy, as well as other aggregated data deriving from the Anti-Money Laundering Aggregate Reports (SARA) by the Bank of Italy's Financial Intelligence Unit (UIF). A joint Bank of Italy-Istat working group was set up to produce new time series from

[13] See https://www.sia.eu/en/media-events/news-press-releases/sia-2018-financial-statement-a-record-year-great-progress-in-card-payments; https://www.nexi.it/investor-relations.html.

these new sources, to study the characteristics of such series and test their adoption in the nowcasting or forecasting of macroeconomic aggregates. Moreover, to foster the participation to the ESSnet Project "Big Data 2018-2020, Bank of Italy (Banca d'Italia) and Istat signed a Memorandum of understanding for mutual collaboration aimed at accessing relevant aggregated FTD held by the Bank of Italy or by other private data-holders and in line with the legal framework described in Section 2.2.1.

BI-COMP and TARGET2 retail database - The joint Bank of Italy-Istat working group focussed on 15 monthly time series of transactions (eight in amount and nine in volume) broken down by payment instruments and reconstructed backwards since January 2000. Microdata of the two settlement systems were aggregated to comply with the General Data Protection Regulation (GDPR). The series are timely but the information has little granularity. Furthermore, it was necessary a pre-treatment and inspection phase before to use the series in order to take into account the changes in the payments regulation that created some shocks during the considered period.

Anti-money laundering aggregate reports (SARA) database - banks and other financial intermediaries have to report[14] monthly to the UIF all transactions amounting to over 15,000 euros, after aggregating them by branch, customer sector and type of transaction. SARA reports do not contain reference to the customers' personal data, but they have information on the customer's sector of economic activity and place of residence. SARA data collection started in January 1993 and, over the years, intervening amendments to the regulatory frameworks affected the reporting scheme (e.g., the thresholds for transactions to be reported and the domains of some classification variables). As some series revealed perturbations, due to changes in the technology used for reporting the series of cash/debit amounts, even in this case a pre-treatment phase was needed to overcome the fluctuations.

A4.2.1.2 Private operators database

During the first quarter of 2019, SIA and NEXI were contacted by the Bank of Italy in order to:

1) identify a possible information set based on aggregations of data already prepared by the managers for internal purposes (at the monthly or daily frequency), grouped by type of technology, or on a socio-economic, sectoral, product or geographical basis;
2) evaluate privacy issues (e.g., consent by banks who are the real owner of card processors' data);
3) define technical methods for collecting, storing and exploiting the data (e.g., periodic data transmission vs. direct remote access to the card processors' data lake).

Bank of Italy is informed on the centralised data collection by the processors in compliance with any profiles of privacy and confidentiality. Bank of Italy should start the realization phase as soon as the pre-treatment and inspection phases are completed, conditional to the solution of a plethora of technical issues such as the analysis of the reporting burden for the card processors, and the construction of a data lake inside the card processors to harmonise different data platforms, etc.

A4.3 Metadata for FTD

Payment system data are timely available with a high frequency patterns. However, such data are not granular in terms of commercial and geographical breakdowns because they come from predefined interbank procedures at the settlement level. The monthly and daily aggregated data we could access are the BI-COMP transactions broken down by payment instruments, namely, credit transfer, direct debit (rav mav, rid, sepa), direct debit payments cards (atm, pos), cheques, other; and the TARGET2 customer payments total and cross-border series.
Referring to the SARA UIF database system, about fifty monthly series, referring to the total value of the banking transactions, cash operations and domestic or foreign wire transfers are considered since the

[14] Pursuant to Article no. 33 of Italian Anti-Money Laundering Law (Legislative Decree no. 231/2007).

beginning of years 2000. Information on the bank branch where the transaction took place and the customer residence (region or foreign country) are also available since 2012. The classification of the sector of economic activity (Total Services, Retail Trade Wholesale Trade Manufacturing Industry Other) has undergone various amendments over the years.

Private Operators

The card processors are sharing with the Bank of Italy a small aggregated dataset for preliminary investigation and testing based on a short string of variables. Daily information allows to geo-localise the direction of the payment flows: the location of the merchant (or Point of Sales) and the location of the issuer (as a proxy of the location of the consumer, if such information is not available), both from the acquiring side (incoming transactions) and the issuing side (outgoing transactions). This kind of breakdown is useful especially for measuring the cross-border flows. Other information also allows the distinction between e-commerce and proximity payments, useful to better measure the retail trade turnover. A valuable piece of information we obtained is the Merchant Category Code (a four-digit code describing the merchant's primary business based on annual sales volume, measured in local currency). About 700 MCCs are provided from the card networks to track the business and the type of goods or services provided. Such codes are similar to the NACE codes, but there is no standard methodology to match the two classifications. However, in the experimental phase of the project, following the card processor need to reduce the computational burden, MCCs codes have been aggregated. Consequently, an accurate preliminary study of the correspondence of the sectors between the MCCs groups of codes and the NACE rev.2 classification groups is needed.

A4.4 Promising official statistics based on FTD

Istat carried out, in January 2019, an internal (across the Directorates) survey has been carried out to highlight the statistical production potentially interested in the use of private operators aggregate data. Seven case studies emerged; four of them refer to Business statistics and National accounts Directorates, the others to Social statistics Directorate. The National accounts Directorate proposed the joint use of FTD and short-term indicators for the early estimates of macroeconomic aggregates and the Economic research Unit suggested the use of FTD to improve the forecasts of private consumption and investment. The Business Short-term statistics Directorate nominated a case study aimed at improving the quality of two monthly indicators (retail trade index and services turnover index) besides a case study on the analysis of the Sharing economy platform characteristics. The Social statistics Directorate proposed the study of the use of the sharing economy platform made by citizens (e.g., for sharing personal assets, for direct peer-to-peer exchange, on-demand services and work, peer to business in the sharing mobility, etc.). Other Social Statistics cases proposed are the study of the e-commerce expenditure and the expenditures for tourism. All pilots envisaged the use of monthly data (and at a lesser extent, yearly data or daily data) and both the value and the number of transactions. The breakdown by type, location where the payments is made, payment channel and industry are also largely requested.
Considering the available information in the FTD managed by the Bank of Italy and the degree of aggregation of information supplied by private providers we could have access, we expect that the statistics that could benefit the most from using FTD are the following:

- *household expenditure of residents for tourism and balance of payments estimates* - available information, from the acquirer side, on the location of the issuing bank of the card in addition to information on the location of the sellers could allow distinguishing the purchases of foreigners in Italy from the purchases of the Italians abroad. These data will be compared with those of the balance of payments and new indicators on the (inbound and outbound) tourism spending could be set on a national basis (and possibly on a regional basis).

- *e-commerce turnover by MCCs groups* - using private credit card operators data a pilot study to estimate the e-commerce turnover will be carried out in order to compare these information with the current e-commerce turnover index and to get some insights on the breakdown by MCCs groups.
- *forecasting activities* - BI-COMP and TARGET2 (monthly and daily) series will be used for forecasting purposes. The availability of private credit card operators FTD for a time horizon of at least 5 years (broken down by type of payment channel and MCCs groups) will allow refining the forecasting of private consumption and retail trade index at national level.
- *early estimates of turnover in Services sector* – monthly time series built with the private operators data (broken down by MCCs groups) will be included in the National accounts model for early estimates in Services sector, to verify their predictive performance.

Appendix 5. Norway: financial transaction data (FTD)

A5.1 The different FTD

Norway has the usual payment instruments, i.e. giro, debit cards, delayed debit cards, credit cards and prepaid cards. For system where data are generated or maintained, however, there are some differences from what we described in Section 2.1.2: Firstly, the National Central bank has a limited role in handling single transactions, c.f. Section A5.2. Secondly, in addition to clearing/settlement systems, card operators/issuers and tax/control authorities, there are the following additional sources in Norway:

- The national payment system for use of debit cards in POS terminals and ATMs.
- The national payment system for giro transfers
- Issuers and acquirers of international payment cards (that can be debit cards, delayed debit cards, credit cards and prepaid cards) and issuers of national credit cards.
- Mobile Payment systems
- DSOP kontroll: A comprehensive structure of FTD made in cooperation between governmental and financial institutions.
- The Register of Crossborder Currency Transactions and Currency Exchange (The Norwegian Tax Administration)

Below we will for all the systems look at the data's relevance for official statistics.

In Norway, the clearance of financial transactions between two bank accounts involving two banks operating in Norway, goes through *NICS, the interbank clearing system of the Norwegian Krone*. This is a pure clearing system that has no role in the settlement. Further, no credit is involved by neither any of the parties nor by NICS.[15] The settlement process is depending on whether a bank is a "level 1" or a "level 2" bank. A "level 1" bank has a settlement account in Norges Bank to make settlements on behalf of themselves and other banks / "level 2" banks. "Level 2" banks have their settlement accounts in a "level 1" bank.

The gross clearance system of NICS involves all transactions that are either exceeding 25 million NOK, marked with REG or HASTE (priority/hurry), or being a SWIFT-transaction received after 11:30.[16]. These micro-transactions are cleared one by one immediately. Gross clearances can only be made by "level 1" banks. Five times each day, multilateral net clearings are calculated within NICS.

All banks can take part in the net clearing system which involves all transactions that are not a part of the gross clearance system above. First the net position of each bank is calculated. As an example, for bank A, in NICS it is calculated how much money bank A should send to or receive from each of the other banks based on the transaction between accounts in these banks and bank A. The transactions here considered, are those that took place during the hours since the previous clearing.

The net position of a level 2 bank is added to the net position of level 2 bank's private settlement bank at level 1. The net position of the private settlement bank, including the underlying level 2 banks' net positions, is sent to Norges Bank for settlement via the level 1 bank's settlement account in the Norwegian central bank settlement system – NBO. Afterwards, each level 1 bank settles the situation for each of their level 2 banks, i.e. moves money between these banks according to the clearance result. Finally, each bank performs a settlement for each of the single bank accounts in that bank.

[15] https://www.bits.no/en/nics/prinsippet-om-kreditering-etter-oppgjor/

[16] https://www.bits.no/document/arsrapport-for-nics-u-vedlegg/ (in Norwegian). SWIFT transactions are mostly (only?) transactions to and from foreign accounts.

Whereas the gross clearance system of NICS has FTD on micro-level between payer and receiver, the net clearance system has only the net transaction between each of the involved banks that will settle the aggregated total between the banks after all the transactions between the banks during the 3-4-hour period since the last clearance was performed.

The settlement system has been studied for the purpose of being a possible source to FTD. However, as we have seen, in Norway the settlement is partly done within the NBO, the National central bank system, and partly within the level 1 banks. For both these sources, the data are similar to that of the clearance system. To get the micro-data transactions hidden behind the net clearances, it would probably be necessary to go to each bank where the internal settlement between the bank accounts are done or to one or more of the bank data centrals handling the data on behalf of banks. However, given necessary historical information about transactions from the payer in bank A all the way to the payee in bank B might not be easily obtained.

The national card payment system of Norway, is a scheme that can be used in ATMs for cash withdrawals and POS terminals for both payments and cash withdrawals ("cash-back"). In 2018, 79 per cent of all payments in POS-terminals in Norway were such payments. Other payment cards can be used in ATMs and POS-terminals as well. Micro-data exists in one database for use of all payment cards in all POS-terminals accepting the national card scheme (almost all terminals) in Norway. This database is a potential source for Official statistics.

The national giro payment system cover both paper-based and electronic credit transfers and direct debits. A major part of these transactions are registered in one database. A considerable part of giro payments are linked to payment orders and electronic invoices . B2C Payment orders exist in one database only, with information about the payer, payee, amount and date of maturity. Electronic Invoices with specifications of the purposes of giro payments are stored in invoice hotels (About 20 B2C invoice hotels) or (for B2B invoices) access points.[17]

DSOP Kontroll is a system that is being constructed as a cooperation project between the financial institutions and three governmental institutions (Tax Authority, the police and the National Insurance Agency (NAV)) for putting less response burden on the financial institutions when getting auditing requests from the governmental institutions. The acronym DSOP is an abbreviation from "Digital cooperation public private". Below we will look more closely into the DSOP Kontroll system.

Today, when the tax authorities perform an audit of a person or an organisation, it contacts the financial institutions. This is a time-consuming process for the financial institutions as well as for the governmental institutions since it is not automated as much as possible.

The system is also helpful for the financial institutions requests to each other, e.g. when a person is applying for a loan, a credit check is made where the banks and other financial institutions are contacted and finds relevant information that is provided to the credit checking company. Receiving such information will be much easier through this digitalised and integrated system.

DSOP Kontroll is a solution to the above challenge. Once operational in 2020, the system will work in the following way:

- First step: Through a simple API-interface, an institution A having legitimate authority to get information, will send a list of personal/organisational identificators to the system. Also an institution having a consent from the persons/organisations in question, will be able to use this system eventually, i.e. accountants hired by a private firm to audit the firm's accounts.

[17] Registered at https://www.anskaffelser.no/verktoy/veiledere/aksesspunkter-ehf-og-bis-formater

- The list is sent to a register, *KAR,* maintained by the finance industry. This is a register that for all persons and organisations, contains have a list of all the 130[18] banks where each person/organisation has an affiliation.
- Automatically, a request is sent from institution A to the banks identified above for getting a list of the account numbers for this person/institution.
- Institution A asks for account summary or transaction history, and this information is then sent automatically.

Notice that the DSOP kontroll system, upon authority to request information, returns potentially all financial transactions for a person/organisation on micro level, at least all bank account to bank account transactions whether being debit card transactions or giro transactions. Statistics Norway has authority based on the Statistical act since this is a register.

The register of transactions crossing the national border, is a register containing information about all cross-border transactions. Bank transactions (SWIFT and currency cheques) exceeding NOK 100 000 (approx. 10 000 euro) and card transactions exceeding 25 000 NOK (approx. 2500 euro) must be reported with detailed information about the payer, the payee, currency and the purpose of the payment. Smaller transactions are available aggregated to receiving country, i.e. for each person or institution sending money abroad.; All transactions from one card holder to e.g. UK are aggregated to one record giving the total amount sent to that country (during the reference time).

A5.2 Getting FTD (Process)

The process of getting FTD in Norway is similar to the general considerations described in Section 2.2, where also the legal situation in Norway was described. Below we will make some additional comments.

The National central bank is a natural source for obtaining information on the financial system as in other countries, but it seems not to be a natural source for getting FTD. It has no role in the clearance process and only a part of the role in the settlement process. The clearance system is preferred over the settlement system which is Norway is distributed among several actors, of which the National central bank is only one of these. Clearance data have been received by Statistics Norway for one day in 2016. More details on these data are given in Section A5.3, where we also will discuss the limitations more in detail.

Debit card data from the National payment system (Section A5.1) covers about 80% of POS payments and about 70% of all card payments in Norway, and is a natural most prioritised source for card data. Such data has been acquired by Statistics Norway for one day in 2016, more details are found in Section A5.3.

Debit and credit card data from international debit cards used in Norway are maintained by seven different acquirers. National credit card data are maintained by a few issuers. Thus, it is a somewhat larger process to obtain data from the use of such cards. However, these data are relevant for Statistics Norway since most payments over the internet as well as contactless payments against POS terminals are made by these cards.

Regarding the possibility the get more data than authorised by the Statistical act, this is impossible at least for microdata. However, the Statistical act enables access to all data needed for the development and production of official statistics. Statistics Norway cannot pay for data and neither for the workload put onto a data provider. However, as mentioned in Section 2.2.2.4, it could be useful both for the data provider as well as for Statistics Norway that an employee in Statistics Norway, for a period of time is working together with the data provider' staff in their office, both for getting insight into the data and to help the data provider preparing data for Statistics Norway.

[18] About 130 banks are already a part of the DSOP Kontroll system a year before it starts to operate.

A5.3 Metadata for FTD

Variables and coverage in the National debit card payment system:

First, we notice that the national debit card payment system data cover most of the debit cards payments and the POS terminal payment platform (c.f. the dimensions of Section 2.1.2) . The relevant variables are:

- total transaction amount (NOK)
- cash withdrawal amount (NOK)
- date of transaction
- time of transaction (HH:MM:SS)
- total amount of purchase (NOK) (this is equal to the total transaction amount in the case when no cash withdrawal has been made, otherwise the difference is reflecting the amount of cash withdrawal).
- account number of payer
- account number of receiver (usually not the final receiver but the interim account of the bank)

The last two variables are pseudonymised in the processing at Statistics Norway.

For each individual transaction, register-information at Statistics Norway on payer (person) and enterprise (receiver) can be linked, using the bank account register from the Tax authorities. Example: demographic variables, household variables, number of employed in enterprise etc. Notice that due to the pseudonymisation above, the data sets used in Statistics Norway has no identification of the person or household involved in the different transactions.

A variable describing the Industry classification code was created, but this is the industry of the enterprise, not of the local unit.

Notice that apart from the payment instrument, platform and total purchase, the variables above are only those needed to identify the pure financial transaction: the transaction amount, the payer, the receiver, and the time of the transaction.

Coverage: The National payment system covers most of the debit card payments in POS terminals. As such it covers almost all of groceries sales except the proportion (roughly 20%) made by cash and the proportion made by international payment cards, domestic credit cards and cards issued by retail chains and used in POS-terminals only (in total 17 per cent of the sales in 2018). For other retail sales types, the proportion will be smaller due to the internet purchases which is often done with international payment cards, and giro, and which is still rather smaller on groceries in Norway than for sports equipment, electronics, books etc.

As seen above, groceries are covered roughly well by the national payment system. In the future, also for groceries there will be more internet sales, which will probably make the national payment system less representative for our purpose.

Contents of the purchase: Payment transactions have little or no information on what has been purchased. However, the stores have purchase transactions (corresponding to the receipts) containing: POS terminal, total purchase amount, and for each item purchased the amount of money and description of item[19]. However, the receipt has no information on who made the purchase.

[19] A relevant tool here is the so-called COICOP codes, "Classification of Individual Consumption According to Purpose". As an example, some of the categories are: "01.1.1.3 Bread and bakery products", "01.1.1.4 Breakfast cereals", "01.1.4.1 Raw and whole milk", "01.1.4.2 Skimmed milk", c.f. https://unstats.un.org/unsd/classifications/business-trade/desc/COICOP_english/COICOP_2018_-_pre-edited_white_cover_version_-_2018-12-26.pdf

Variables and coverage in the NICS system:

The NICS system data has the pure financial transaction identification variables: payer (account), receiver (account), amount and time. These are the interesting variables for our purpose in this system. For giro transactions, there might in some cases be overall information on what the transaction is about, but this seems only to be the text field filled in for the giro, e.g. manually by the payer, and is only meant as notes for the payer and receiver within the context of their relation. Thus, it is difficult to classify the information into meaningful categories such as "payment for hiring an apartment/house". As an example, the text could just be the name of the city or country – information being precise for both parties.

Since the NICS system is the output from the clearance system which has both net clearing and gross clearing, the only relevant FTD for Statistics Norway, are from the gross clearing system, containing transactions exceeding 25 million NOK (about 2.5 million euros), in addition to lower value transactions marked "REG", "HASTE" (urgent/priority), or being SWIFT transactions received after 11:30, c.f. Section A5.1. Thus, for most of the transactions, a micro-level transaction is only the sum of all transactions between two certain banks made during the corresponding time interval (about 3-4 hours) of the day.

For the population of interest, i.e. debit cards, credit cards and giro transactions, we here have both debit cards and giro, but we only cover the part of the population cleared by the gross clearance system, which we can assume to be systematically different from the transactions hidden behind the net clearance system, at least when it comes to amount of money, but possibly also when it comes to purposes of the payments.

Variables and coverage in the DSOP system

We do not yet know which variables will be contained in the DSOP system, but it should at least be the pure financial transaction variables (payer, receiver, amount, time and type of account). Supposedly, the payment instrument or the type of payment should be identified. Type of payment at a specific payment platform must perhaps be identified by elimination process: those card transactions not being payments in POS terminals, are mostly made on web or as mobile payments

The DSOP kontroll system is covering information connected to the use of debit cards, cheque and giro except those for a few banks that are not yet included. A challenge will be lack of information connected to the use of credit cards and mobile payments and technical difficulties of requesting the whole population or larger parts of the population from the system. Fortunately, there is as solution to this challenge: a random sample of units (persons/organisations) can be drawn by Statistics Norway and a request can be sent to the system for only the sample. Then, estimates can be made in the usual way for statistics based on sample surveys.

Variables and coverage in the register of transactions crossing the national border

The register of transactions crossing the national border is as the name tells, not covering transactions within Norway, but it is covering all cross-border transactions by use of international payment cards, SWIFT bank to bank transfers (in Norway often initiated by use of the national giro system) as well as other cross border transactions, i.e. transactions effectuated by entities registered as payment institutions

Although covering all domestic persons and organizations for the cross-border transactions mentioned above, the register has partly aggregated data as we noticed in Section A5.1. In more detail: The information for bank transactions exceeding 100 000 NOK and card transactions exceeding 25000 NOK is about the person/organisation id of payer, payee, country, currency and the purpose of the transaction, and total amount. Bank transactions less than 100 000 NOK lack information about the purposes of the payments. When a card transaction is less than 25000 NOK, all transactions with a person/organisation's card are aggregated to monthly numbers for a certain country where the card has been used (for outgoing transactions or (for ingoing transactions) the country where the card is issued. For ingoing card transactions

less than 25 000 NOK, the banks also send a monthly report with aggregated numbers and amounts of payments for every single user location, with detailed information about the user location. Thus, from this data, it should be possible to construct a total overview of the use of card issued abroad on user locations in Norway.

A5.4 Official statistics potentially benefitting from FTD

A5.4.1 Suggested statistics reported from within the NSI

Internal surveys within Statistics Norway has been performed. In which official statistics is it foreseen a potential improvement by using Financial transaction data? Different usages have been pointed out:

- In foreign trade statistics: flows of payments across the border associated with Norwegian companies
- Statistics on payment instruments.
- In statistics in border trade: amount of purchases abroad for inhabitants of Norway and purchases in Norway made by foreigners/tourists.
- Consumer expenditure survey: Consumption (detailed)
- Consumer price index: price development of the purchases made by households
- Structural statistics on wholesale and retail trade: the activity in retail sales industry
- Retail sales index: value and volume of retail sales
- Index of monthly turnover: turnover in the construction industry and the service industry.
- Construction cost indexes: price of the assets used in road construction and construction of new dwellings.
- Energy statistics: energy consumption expenditure by household type.
- National accounts on retail sale.

A5.4.2 The promising statistics

Below is a list of promising statistics, either as an additional source for improving the statistics or quality evaluating the statistics, or as a source that can replace the existing source.

- Retails sales index is a promising statistics, since potentially the current monthly data collection from individual stores could be replaced by aggregated data from the debit card national payment system, together with aggregated purchase transaction data from the grocery stores.
    - For the grocery store sector it is then possible to see the total sales from the purchase data, and then calculate the proportion of sales performed with the debit card instrument.
    - Outside the grocery store sector, the debit cards total sales is available. Then by some modelling based on the grocery store sector, the total sales from all payment instruments can be estimated. The result will rely on strong modelling assumptions, and it might not be possible to replace the current source completely but perhaps partially. In any case, FTD can be used as an alternative source that might be used either in the estimation or in quality assessment of the current statistics.
- Statistics on economic relations between companies and industries**:** based on giro transactions, the transactions could be aggregated so that each record/line in a dataset would be the sum of transactions between two companies during e.g. a year. The (composite) unit is then the pair (payer, receiver). Then it would be possible to use the NICS data to quantify the amount of transactions between e.g. different industries at the chosen level of industry classification code. This would give a valuable insight into the business structure since it shows which industries is trading with each other. The clearance data could be used, but due to the aggregation property (Section A5.1), DSOP Kontroll is a source more likely to be successful.

Appendix 6. Portugal: financial transaction data (FTD)

A6.1 The different FTD

A6.1.1 Types of FTD

In Portugal FTD can be accessed in different ways, each of them covering slightly different domains. The universe of Financial Transaction Data includes electronic and non-electronic transactions. Among electronic transactions we may distinguish from four main types of FTD:

- MBWAY and MULTIBANCO (electronic payment and money transfer processes managed by SIBS, a payment services provider). SIBS is also responsible for automatic payment terminals (POS);
- Electronic transactions managed by Credit Card companies (we assume they are mostly electronic nowadays);
- peer–to-peer electronic transfers between banks.
- e-invoice

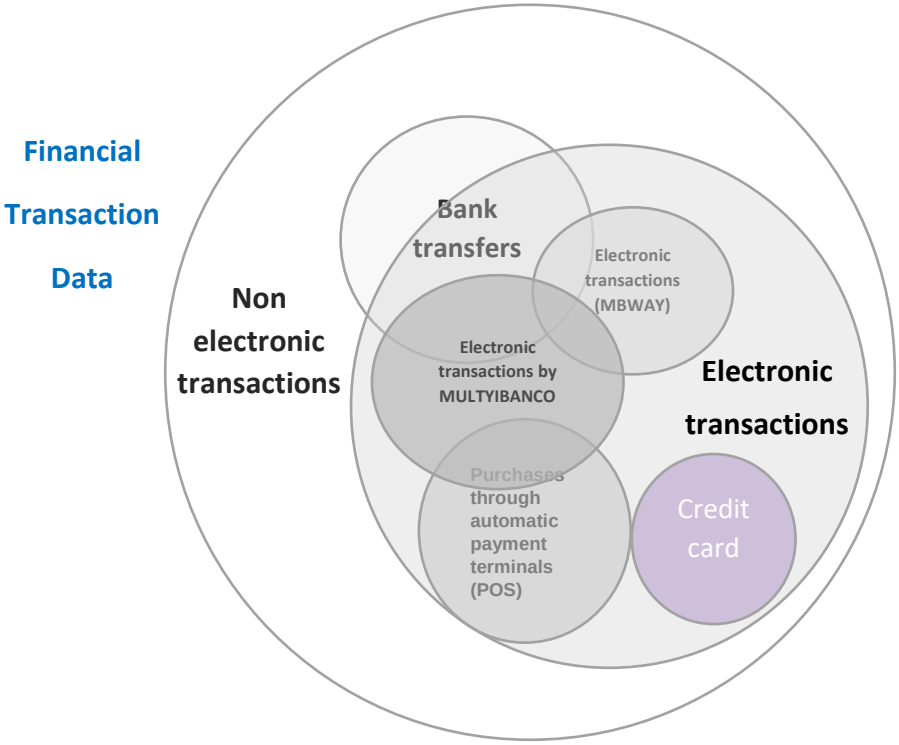


Figure A 6.1: Financial Transaction Data in Portugal

Electronic transactions (MBWAY) are mainly mobile phone transactions, MULTIBANCO are mainly desktop, notebook and ATM transactions. Purchases through automatic payment terminals (POS)

FTD are, then, all the universe of financial transactions being electronic or non-electronic transactions, and that may use an electronic system (POS terminals, ATM MULTIBANCO, or MBWAY) or not (simple bank transfer). In the latter case, we refer to peer-to-peer bank transfer where electronic system may be needed (or not) as intermediary. FTD from e-Invoice are related to all transactions (electronic and non-electronic) collected by the Tax Authority. This type of data covers all FTD corresponding to the whole universe depicted in Figure A6.1.

A6.1.2 Data owners and data providers

The data owners consist of the private companies, and banks, and the Bank of Portugal. The access to data from private companies and banks is difficult since these companies are not obligated by law to report microdata to Statistics Portugal. The Bank of Portugal collects this information for the transactions data in aggregated way, and SIBS (who manages MBWAY and MULTIBNCO), manages almost all electronic transactions (over 60% or all transactions are made electronically and almost all are monitored by SIBS).

SIBS is a leading payment services provider in various countries in Europe and Africa. The company is responsible for the ATM Express network and MULTIBANCO which include ATM machines, automatic payment terminals and mobile and online alternatives. Recently, a new system has been created, MBWAY, a framework developed for mobile phones.

Financial transactions or services covered by SIBS are the following:

- Card business

  SIBS is responsible for the emission and processing of debit, credit or pre-paid cards from national and international brands.

- Payment services

  Management of services of interbank transfers (credit transfers, direct debits) in SEPA space.

- ATM and POS terminals

  SIBS manages and monitors ATM terminal networks (MULTIBANCO or others). It also supervises POS network.

In Portugal, by law, all the purchases are associated with invoices. Financial transactions that correspond to purchases of goods or services are also monitored by the Tax Authority, that keeps track of all the FTD that correspond to the issue of an invoice. The E-invoice is a dataset of the Tax Authority to monitor all the electronic invoices.

Therefore, the data providers we may consider are:

Data provider	Data
SIBS, Interbank Settlement of the System of Payments	Electronic transactions (payments) made by ATM in MULTIBANCO and MBWAY and purchases through POS
Bank of Portugal	Direct debits and transfers, payments, withdrawals and purchases
Tax Authority	FTD from E-Invoice

A6.2 Getting FTD

A6.2.1 Legal requirements for FTD

Statistical Act

The Portuguese National Statistical System (NSS) includes the Statistical Council, the State body that superintends and coordinates the system; the National Statistical Institute, IP (Statistics Portugal), the central body responsible for the production and dissemination of official statistics, that ensures the supervision and the technical and scientific coordination of the NSS; the Bank of Portugal that, as part of its mission, is responsible for the collection and compilation of monetary, financial, foreign exchange and balance of

payments statistics; the Regional Services of Statistics of the Autonomous Regions of Açores and Madeira that act as delegations of the Statistics Portugal in relation to nationwide statistics as statistical authorities in what concerns regional statistics; and other entities producing official statistics by delegation of Statistics Portugal.

Statistics Portugal, Bank of Portugal, Regional Services of Statistics of the Autonomous Regions of Açores and Madeira and entities producing official statistics by delegation of Statistics Portugal are considered statistical authorities, having responsibility for the production of official statistics, and empowered to require (mandatory and gratuitously) to all departments or agencies, individuals and legal entities, information necessary for the production of official statistics.

The official statistics are produced with technical independence and considered as public good, observing the national and international quality standards, and meeting the users' needs in an efficient manner, preventing the providers of information to the statistical authorities from an excessive burden, by using increasingly the administrative data.

All personal data collected by the statistical authorities for statistical purposes, are considered confidential, and legally protected. All people connected with the production of official statistics are obliged to professional secrecy. To break confidentiality is considered a very serious administrative offence that implies criminal responsibility.

The entities responsible for the production of official statistics have the power to access any service or organism, single or plural person in order to gather any element necessary to the production of official statistics. However, there may be limitations to this access (article 4) and obtaining information can be dismissed if it appears impossible or of disproportional efforts.

A6.2.2 Process for getting FTD

The different transactions, B2B, B2C through bank cards, direct debits and transfers, payments, withdrawals and purchases are all monitored by the Bank of Portugal and the Tax Authority.
The Bank of Portugal publishes a variety of aggregated statistics concerning the Portuguese economy as public finances, financial national accounts, the banking system, the payment system, financial and monetary statistics amongst others.

The information provided by the Bank of Portugal is not detailed and there is no current agreement between Statistics Portugal and Bank of Portugal that facilitates the access to this data.

SIBS also detains many financial transaction data since it is responsible for processing transactions with various banking payment instruments. Although it is a private company, Statistics Portugal already has access to some data by SIBS in an aggregated way. Recently, contacts have been made between Statistics Portugal in order to explore new forms of interchanging data.

FTD registered in the Tax authority are currently the best data to be used in the ESSnet Big Data. A protocol between Statistics Portugal and the Tax Authority has been signed last year that allows the access of Statistics Portugal to data from E-Invoice data set.

Nonetheless, we will try to continue to contact SIBS and Bank of Portugal.

In case we are unable to make an agreement with any of these entities, as mentioned before, Statistics Portugal will use the aggregated data from SIBS, namely the withdrawals from ATM machines and the purchases through automatic payment terminals. We will try to draw results from this data, which is a proxy of financial transaction data for internal demand.

A6.3 Metadata for FTD

Relevant topics that should be considered for each FTD dataset:

E-invoice contains the following information for each register (FTD):

- VAT number (taxpayer number)
- VAT of the buyer
- Month of transaction
- Country of the buyer
- Taxable value (aggregated by VAT number)

Regarding the other data providers, we do not know what kind of metadata we would have access to.

A6.4 Official statistics potentially benefitting from FTD

A6.4.1 Suggested statistics reported from within the NSI

(This subsection is only relevant if experts on the different official statistics at the NSI have been asked what potential they see in FTD)

Depending on the access to the E-Invoice dataset, Statistics Portugal may use this data to produce flash estimates for macroeconomic indicators and private consumption estimation. Furthermore, this information is important to measure Collaborative Economy, namely to estimate the business volume in specific activity sectors. By having access to the identifiers of the companies in some domains (tourism, transports, etc.), we may estimate the aggregate turnover in these sectors.

A6.4.2 The promising statistics

Below is a list of promising statistics, either as an additional source for improving the statistics or quality evaluating the statistics, or as a source that can replace the existing source. For each statistics, it is indicated which FTD that should be used (and possibly which variable in the FTD dataset).

Statistics	**FTD**	**Source**	**Variable**
Turnover of collaborative Economy (total)	E-Invoice	Tax Authority	all
Turnover of collaborative Economy (specific sector)	E-Invoice	Tax Authority	all
Internal demand	Withdrawals and payments	SIBS	

Appendix 7. Slovenia: financial transaction data (FTD)

A7.1 The different FTD

Types of FTD that could possibly be obtained:

- Payment cards (debit cards and credit cards)
- Cash withdrawals and deposits
- Electronic banking
- Credit transfers
- Direct debits (giro payment)
- Fiscal cash registers data
- e-invoice data (In Slovenia e-invoice is only used by the government -- for direct budget users)

A7.1.1 FTD controlled by the Central bank

Since organizations that record electronic payments with payment or credit cards are subject to Slovenian national law, all financial transactions performed between consumers and businesses (B2C) and between businesses (B2B), both debit and credit card and giro transactions, are legally monitored and recorded by Bank of Slovenia.

There are currently 18 financial institutions in Slovenia, which according to the law are obliged to report the data to Bank of Slovenia:

- Addiko bank d.d.
- BKS BANK AG, Bančna podružnica
- Banka Sparkasse d.d.
- RCI Banque Societe Anonyme, podr
- Abanka d.d.
- SKB banka d.d.
- Banka Intesa Sanpaolo d.d.
- Gorenjska banka d.d., Kranj
- Primorska hranilnica d.d.
- DBS d.d. Ljubljana
- UniCredit Banka Slovenija d.d.
- Delavska hranilnica d.d.
- Sberbank banka d.d.
- Hranilnica LON d.d. Kranj
- SID banka d.d., Ljubljana
- Nova LB d.d. Ljubljana
- Nova Kreditna banka Maribor d.d.
- MBILLS d.o.o.

Bank of Slovenia publishes information on payment cards, withdrawals and deposits in cash, electronic banking, credit transfers and direct debits. Data are published quarterly, and are available from 2015 onwards. Below is a more detailed breakdown of data collected and published by the BS.

Payment cards

Data on payment cards are shown by type: credit cards, deferred payment cards, debit cards, and cards with an e-money function. The latter is similar to prepaid cards. Because only one bank issues cards with e-money function in Slovenia, these transactions are added to debit cards in order not to disclose individual data. The

number and value of payments is broken down according to whether payments were made at home or abroad. Payments with foreign cards in Slovenia are also shown.

Table A7.1: Number of payment cards issued by resident issuers

Payment cards - Total	Debit cards and cards with e-money function	Delayed debit cards	Credit cards

Table A7.2: Volume of card payments

Payment cards issued by resident issuers						Payment cards issued by non-resident issuers
Debit cards and cards with e-money function		Delayed debit cards		Credit cards		
in Slovenia	cross-border	in Slovenia	cross-border	in Slovenia	cross-border	in Slovenia

Cash withdrawals and deposits

Bank of Slovenia publishes data on withdrawals and deposits at ATMs in Slovenia, withdrawals with Slovenian cards at ATMs abroad and withdrawals and deposits at bank counters in Slovenia.

Table A7.3: ATMs in Slovenia - transactions with cards issued by resident and non-resident issuers

Number of ATMs	Withdrawals						Deposits	
	Volume			Value in mio EUR			Volume	Value in mio EUR
	with cards issued by resident issuers - on us	with cards issued by resident issuers - not on us	with cards issued by non-resident issuers	with cards issued by resident issuers - on us	with cards issued by resident issuers - not on us	with cards issued by non-resident issuers	with cards issued by resident issuers	with cards issued by resident issuers

Table A7.4: ATMs abroad - cash withdrawals with cards issued by resident issuers

Withdrawals	
Volume	Value in mio EUR

Table A7.5: Over The Counter cash withdrawals and cash deposits in Slovenia

Withdrawals		Deposits	
Volume	Value in mio EUR	Volume	Value in mio EUR

Electronic banking

Electronic banking data are published for online banking, telephone banking and mobile banking in the form of a breakdown in Table A7.6.

Table A7.6: Electronic banking

Number of users		Volume of payments		Value of payments in mio EUR	
Natural persons, Sole proprietors	Legal persons	in Slovenia	cross-border	in Slovenia	cross-border

Credit transfers

Credit transfers data for domestic transactions, cross-border transactions and all transactions together are published in the form of the breakdown below.

Table A7.5: Credit transfers by volume and value

Credit transfers - volume				Credit transfers - value in mio EUR			
initiated in a paper-based form	initiated electronically			initiated in a paper-based form	initiated electronically		
		initiated ina file/batch	initiated on a single payment basis			initiated in a file/batch	initiated on a single payment basis

Direct debits

The direct debit data for domestic transactions, cross-border transactions and all transactions together are published in the form of the breakdown below.

Table A7.6: Direct debits by volume and value

Direct debits - volume		Direct debits - value in mio EUR	
initiated in a file/batch	initiated on a single payment basis	initiated in a file/batch	initiated on a single payment basis

A7.1.2 FTD controlled by private operators

BANKART is a company founded in 1997 by the 22 Slovenian banks on the initiative of Nova Ljubljanska banka, SKB Banka and Abanka for the processing of modern payment instruments. The main mission of the company is to ensure reliable, safe and cost-effective processing of transactions with various banking payment instruments and is almost 100% owned by banks.

Business areas that Bankart covers:

• Processing of ATM operations

Bankart supervises and manages a network of ATMs for commercial banks and savings banks active in retail banking in Slovenia and covers a 93% market share.

• Processing card business

As at 31 December 2017, there were 1,273,461 credit cards and 2,628,499 debit cards in Slovenia - a total of 3,901,960 cards. Bankart processes 628,567 credit and 2,157,767 debit cards, representing 49% of the market share in the field of credit card processing and 82% of the market share in debit cards in Slovenia.

According to the Bank of Slovenia data, as at 31 December 2017, there were 35,842 POS terminals in Slovenia. Bankart processes for 26,674 POS terminals, which represents a 74% market share in Slovenia.

• Payment systems and services

The business area of payment systems and services comprises the management of payment systems and systems for processing SEPA transactions and the E-invoice system. Both SEPA and the SIMP-PS payment system enable the execution of credit payments and direct debits in accordance with SEPA schemes. Each system provides participants with access to all SEPA banks, which means that transactions are executed

between SEPA participants and other euro area banks. The E-invoice system enables the exchange of invoices and other documents in electronic format among the participants.

• ATM and POS terminals

Bankart manages and maintains ATMs and POS network both in terms of daily operational operations and in terms of ensuring compliance with MasterCard and Visa mandates.

A7.1.3 FTD controlled by Financial Administration of the Republic of Slovenia

Another possible source of financial transaction data are data on fiscal cash registers. With fiscal cash register Financial Administration of the Republic of Slovenia monitors all issued invoices, which are paid in money. This includes all payments except credit transfers, meaning also payments with credit or debit cards are included. For the payment cards, both usage in store and online are included.

Not every country uses fiscal money register so there is no uniform definition for it like for example with SEPA . These are systems and models that ensure that data on issued invoices and the turnover of taxable persons reflects the actual economic situation. It is one of the adopted measures against grey economy and tax evasion.

The taxpayers are connected via the internet connection to the information system of the Financial Administration of the Republic of Slovenia. The financial administration confirms and stores data on cash invoices in the process of issuing them in real time. Thus, upon each delivery of goods or services for payment, a taxable person must issue an invoice through an electronic device or using a bound book of accounts, approve it and deliver it to the buyer.

The purpose of carrying out the procedure for the tax invoicing is to obtain a unique identification code of the invoice from the tax authority for each account paid in cash representing a certificate that the turnover generated in cash operations has been reported to the tax authority in the actual scope and cannot therefore be retrofitted with the purpose evading tax.

The invoice (in addition to classic data) must include the time of issue of the invoice (hour and minute), the code of the person/company issuing the invoice using an electronic invoicing device (the name of the person issuing the invoice must be associated with the tax number of that person) , a unique invoice identifier and the security code of the issuer of the invoice.

The invoice number must consist of three parts:

- labels of the business premises of issuer;
- codes of an electronic device for issuing invoices and
- consecutive invoice numbers.

By the law a natural person who carries out activities does not have to pay for the goods supplied and the services rendered to the transaction account of a legal person or natural person performing an activity if the individual payment does not exceed 420 EUR. Likewise, the legal person of payment for the goods supplied and services rendered by him shall not be liable to transfer to the transaction account of the legal or natural person performing the activity if the individual payment does not exceed EUR 420.

Persons engaged in the activity of selling goods or providing services in the Republic of Slovenia may not receive payment in cash from the customer or from any third party in the sale of certain goods or the provision of a particular service, provided that this exceeds the value of EUR 5,000.

A7.2 Getting FTD

A7.2.1 Process for getting FTD

Since all financial transactions are legally monitored and recorded by Bank of Slovenia, data on financial transactions exist, only the access to them is questionable. At present, at national level, the number and value of financial transactions in the framework of the monitoring of payment statistics is monitored by the Bank of Slovenia.

Bank of Slovenia is willing to share the data with the Statistical Office of Slovenia, but according to the current legislation and demands from ECB, banks are obliged to report aggregated quarterly data, within ten days after the completed quarter. In such form, financial transaction data do not represent added value for the Statistical Office of Slovenia, and it would not be sensible to monitor them. At the same time Bank of Slovenia has not expressed willingness to cooperate any further within this project. Nevertheless, the published data can help us, as we can see different types of data that could be obtained.

One possibility of accessing financial transaction data, is also to try to get them from the banks themselves. At the end of year 2018 there were 18 financial institutions in Slovenia, which according to the law are obliged to report Bank of Slovenia data and as such represent a potential source of data.

Since this is a time-consuming process and, in order to increase market share coverage, it would be necessary to obtain data from a large number of banks, it would make sense to look for a third party that would have access to a wider network of banks.

Possible solution is already mentioned privately owned company Bankart that is in charge of processing of transactions with various banking payment instruments. Currently we are in talks with Bankart regarding the data, but at this stage, we have not yet agreed on obtaining the data and as such, we cannot predict if we will be able to access any data before the end on the project ESSnet BD II. The granularity has not been decided, but it will not be microdata.

Another potential source is data on Fiscal cash registers. Based on our talks with Financial Administration we will most likely be able to obtain the data on fiscal cash registers, but we do not know yet, when that will occur. At this stage, we were given an insight into structure of data, we could obtain.

A7.3 Metadata for FTD

As currently we are most likely to obtain the data on fiscal cash registers, this chapter will emphasise on those data.

- **Population**: all issued invoices, which are paid in cash, credit and debit cards, and do not exceed value of EUR 5,000 for consumers (B2C) and EUR 420 for businesses (B2B).
- **Base unit**: individual cash or card payment
- **Composite units** that could be formed: enterprise
- **Variables:**
    - the tax number of the taxpayer who issues the invoice
    - time of issue of the invoice;
    - date of issue of the invoice;
    - the invoice number and method of assigning an invoice number;
    - the value of the invoice, the total value of the base by type of tax, broken down by tax rates;
    - value for payment (EUR);
    - the tax number of the natural person issuing an invoice using an electronic device (actual cashier);
    - the security code of the issuer of the invoice;

- the tax number or identification number for the purpose of the value added tax value of the buyer or the contracting authority in cases where these data are indicated in the account in accordance with the tax regulations;
- number of the original invoice in the case of carrying out the process of confirming the subsequent change of data in the account.
- the code of the obligor's business premises;
- the identification code of the building or part of the building where the business premises of the taxpayer are located, as determined in the real estate register (number of the cadastral municipality, building number and part number of the building);
- the address of the business premises of the taxpayer;
- the type of business premises of the taxpayer when the debtor issues invoices in a movable business premises.

There is no information available about the payer.

All units have an identifier (tax number of enterprise) that can be used at the Statistics of Slovenia to link the data with existing internal data obtained from both surveys and administrative sources.

A7.4 Official statistics potentially benefitting from FTD

A7.4.1 Suggested statistics reported from within the NSI

To recognise areas with potential to use FTD we designed a questionnaire and sent it to experts on the different official statistics at our NSI. Based on the answers the possible need for FTD was detected within following areas:

- Business statistics: foreign trade, retail trade, wholesale, services, industry
- Demographic and social statistics: household consumption, income and living conditions, income and poverty of the population / households, satellite accounts for education, expenditure on formal education, culture statistics
- Environmental statistics: use of ICT, foreign travellers / tourists, travels of domestic population
- Macroeconomic statistics: quarterly GDP, investments, non-financial sector accounts, final consumption of households, PPP and ICES / HICP

As at the moment we do not have any data, we tried to recognise type of data that would be most needed. Based on the questionnaire the most useful data would be debit and credit card data, followed by electronic banking data (number of users, number of payments and value of payments in Slovenia and abroad) and direct debits (number and value of direct debits for domestic and cross-border transactions). Credit transfers and withdrawals and cash deposits would also be useful for some statistics.

All potential users are interested in the value of transactions and in the category of products and services to which the payment relates. More than half of them indicated that they would also need information about the recipient's or payer's activity sector and the location of the payee.

When asked about importance, information on the value of transactions was the most important data, followed by data on the category of products / services to which the payment relates, and the activity of the payer or the recipient.

All of potential users would need microdata more than aggregates, and most of them would use FTD as an additional data source for already existing statistics, or to increase the quality of existing statistics. Others would like to replace existing resources or develop new statistics.

A7.4.2 The promising statistics

Based on the data we are most likely to obtain in time scope of this project, the most promising statistics are statistics on retail trade.

Currently indices on retail trade are calculated based on data from survey and administrative data. Observation units that answer the questionnaires are determined on the basis of their previous turnover in two steps: at the level of the survey and at the level of the activity group. The units are sorted by descending turnover, and then we select a sufficient number of units from the begging of the list to exceed the defined share (approximately 60%) of turnover in the total turnover of units covered in the selected activity group. For the remaining observation units we use administrative data being reported by enterprises to the Financial Administration of the Republic of Slovenia for the value added tax purpose (DDV-O forms).

Administrative data on value added tax, that we are currently using, becomes available to us 45 days after reference period. As the data on cash registers would be available 10 days after reference period at the latest, we could get more accurate results at time of first publishing of provisional data, which is 30 days after reference period.

Based on analysis we could determine whether quality of data on cash registers is sufficient to be used in regular production of Index of turnover in retail trade.

Appendix 8. What is an operational definition

The purpose of literature review aims to define what sharing economy is in an operational way. Operational definition suggests quantitative measures of sharing economy: output, employees, investments, etc. Wikipedia provides the following example of operational definition: *weight is the numbers that appeared when that object in placed on a weighting scale.*

Operational definition differs from purely theoretical definitions, which suggested a way to think about the object. Measurement is an operationalization of theoretical insights. Let us look at two examples from National Accounts that demonstrate difference between operational and "scientific" definition.

- Production boundary definition is fundamental for measuring output of an economy. Production boundary, as defined in the ESA 2010, excludes services produced and consumed within the same household: cleaning, cooking, care of children, old people, sick people, etc. The same holds for volunteer services. ESA 2010 does not includes volunteer services (e.g., cleaning with no payment) in the production boundary and hence services provided are not output.

Economic theory considers both the efforts of households and volunteer as production activity which produces output. The ESA 2010 definition about production boundary shrinks the economic concept of production in order to overcome difficulties when measure the output of households. It is hard to measure the services provided by the household's members and consumed within the same households. We could add it is hard to measure the volume of services provided by one household to another household for free, e.g., sharing households' assets with no middleman between.

When a household contracts someone to clean, cook, etc. and pays for it services provided by contractor fall into production boundary and are part of households' output. There are a contract, payments, social security number, and so on. Statisticians have enough sources of info to measure the output even using administrative data sources.

- ESA 2010 measures a good part of general government sector output as a sum of inputs. For example, the output of a hospital is a sum of expenditures on wages and salaries, medicines, water, electricity, and so on. This way of evaluation of output as a sum of inputs is at odds with economic theory. Production function, which is at the heart of the neoclassical economics links inputs with outputs and built around that linkage a lot of theory. Moreover, evaluation of output as a sum of inputs make comparisons across countries with different size of government sector highly suspicious.

The above example show that ESA definition of production boundary and output is narrow than what economic theory understands about production. ESA 2010 definitions are operational definition, which took into account both the economic logic of phenomena and the ability of national statistics to measure different forms of economic activity. Operational definition reflects limitation in ability to measure.

System of national accounts develop over time and there is a tendency to come closer to economic theory concept of production boundary, output, income, etc. 1953 edition of SNA under auspices of UNSC had about 50 pages. ESA 2010 is as thick is 651 pages. SNA refine concepts and measurement methods and number of pages increases accordingly.

Appendix 9. Questionnaire of the WPG mini-survey on SEP-related indicators

This questionnaire is aimed at assessing how much information available in WPG countries on FTDs could be used also to compare the potential for SEP-related business at national, or sub-national, level.
It focuses only on those FTD which refer to credit card transactions and, among them, to those credit card transactions which are managed on-line, on the Web (whether or not genuine e-commerce transactions).
In this respect, questions 1.1, 1.2 and 1.3 ask about credit cards' FTD in general, while the following questions will be mainly about on-line FTDs.

Essnet Big Data 2 WPG SEP **Questionnaire on the availability of credit card FTD data to estimate the role of digital platforms in EU countries**	
Country:	
Respondent:	
Date:	

Section 1	FTD data source availability		
1.1 Please, list the sources of FTD data (maintainers) which are actual/future data providers for data that will be received within this Essnet WPG and what is their rate of coverage of national FTD total			
Name (or just a code if name cannot be disclosed)	Status (public vs. private)	Coverage (national vs. regional)	Percentage of national total (approx.)
		National total	100%

1.2 As a result of the planned data collection, do you will be able to get any aggregated TOTAL with reference to the volume of FTD credit card data for any period of time (from 1 day to 1 year, either on issuing or acquiring side)?	
Yes ☐	No ☐

1.3 Please, describe below the broadest coverage of FTD credit card data which could be available in your country by two dimensions: time and territory (country/region(s)).	
Time:	
Territory:	
The above coverage can be achieved:	Using data from a single source ☐ Combining more sources ☐
Available aggregated data refer to:	Transactions ☐ Value ☐ Both ☐

1.4 With reference to the above-referred "broadest coverage", do you will be able to make a distinction between credit card FTDs as a result of an on-site (proximity) purchases and credit card FTDs from online transactions (e-commerce)?		
Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
If the answer is "No", please go to Section 3, otherwise go to question 1.5		

1.5 Please, describe below the broadest coverage of online credit card FTD data which could be available in your country by two dimensions: time and area (country/region(s)).	
Time:	
Area: (e.g. "whole country", "only the region ....")	
The above coverage can be achieved:	Using data from a single source ☐ Combining more sources ☐
Available aggregated data refer to:	Transactions ☐ Value ☐ Both ☐

Section 2	Data breakdown		
2.1 With reference to the broadest available coverage of online credit card FTD data (Q.1.5), please report whether a further breakdown could be made available by (either on issuing or acquiring side):			
2.1.1 Payer			
2.1.1.1 Nature (Individual/Firms/Other)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.1.2 Nationality 1 (domestic/foreign)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.1.3 Nationality 2 (country where the card has been issued	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2 Receiver			
2.1.2.1 Nature (Individual/Firms/Other)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2.2 Nationality 1 (domestic/foreign)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2.3 Nationality 2 (country where the card has been issued	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.2.4 Sector of activity (if a firm)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:
2.1.3 Transaction			
2.1.3.1 Value	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:

2.1.3.2 Domain (at least making a distinction between goods and service selling)	Yes ☐	No ☐	Yes, under some constrains ☐ please, specify:

2.2 Please, specify whether some of the backgrounds listed in Q 2.1 can be combined with reference to the national total of FTDs from online transactions.				
2.2.1 Transactions by value				
	Receiver's nature	Receiver's nationality 1	Receiver's nationality 2	Receiver's activity
Payer's nature	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
2.2.2 Transactions by domain				
	Receiver's nature	Receiver's nationality 1	Receiver's nationality 2	Receiver's activity
Payer's nature	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐
Payer's nationality 1	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐	Yes ☐ No ☐

2.3 As a result of the information provided in Q 2.1 and Q 2.2, please describe the main aggregated indicators (or tables) which could be made available.				
	Description	Estimated national coverage (%) of FTDs from online transactions	Number of cells	Reference period
Indicator 1				
Indicator 2				
Indicator 3				
Indicator 4				
Indicator 5				
Indicator 6				

Section 3.	Complementary information	
3.1 In order to complement or to replace a distinction between "proximity" and online FTDs, can other methods be implemented (e.g. by nature of the receiver/merchant)?	Yes ☐ please, specify:	No ☐
3.2 Is it any distinction between "desktop" and "mobile" online FTDs available in your country?	Yes ☐ please, specify:	No ☐
3.3 Is there any way, in your country, to assess the relevance (such as a concentration rate, both in terms of number of transactions and value) of the receivers with the highest number of incoming online payments?	Yes ☐ please, specify:	No ☐
3.4 The same as Q 3.2, by country?	Yes ☐ please, specify:	No ☐
3.5 With reference to Q. 2.1, please describe, if any, the breakdown used to describe the nature of payers/receivers		

3.6 With reference to Q. 2.1, please describe, if any, the breakdown used to describe the nationality of payers/receivers			
3.7 With reference to Q. 2.1.2.4, please describe, if any, the breakdown used to describe the sector of activity of receivers			
3.8 With reference to Q. 2.1.3.2, please describe, if any, the breakdown used to describe the domain of transactions			
3.9 With reference to the indicators described in Q. 2.3, which frequency is expected for the release of new FTD data?	Yearly ☐	Monthly ☐	Other ☐
3.10 In addition to the standard use of FTDs aggregated data, is it any use of FTDs (anonymised) microdata envisaged (e.g. to calculate ratios, to identify clusters/patterns of use, etc.)?	Yes ☐ please, specify:	No ☐	
3.11 Is it any indicator on the "transaction time" (peak, average, etc.), that is the time needed to complete a financial transaction, available in your country?	Yes ☐ please, specify:	No ☐	
3.12 Which quality assessment methods are going to be implemented to check for accuracy and representativeness of the FTDs' sample used). Please, describe if any.			

Appendix 10. Results of the WPG mini-survey

Table A10.1 Results of the WPG mini-survey

	Germany	Norway	Bulgaria	Italy
1.2	Y	Y	Y	Y
1.3	Annual National Single source T+V	Q/M National 2 sources T+V	Annual National Single source V	Annual National Single source T+V
1.4	N	Y, quarterly	Y	Y
1.5	-	Single source T+V	Single source V	Single source T+V
2.1.1.1	-	Only private cards vs. business cards	Not available for non-residents	No
2.1.1.2	-	No	Y	Y
2.1.1.3	-	Y	Not available for non-residents	Y
2.1.2.1	-	From a single source via MCC codes	Not available for non-residents	No
2.1.2.2	-	Y	Y	Y
2.1.2.3	-	Y	Not available for non-residents	Only for selected countries
2.1.2.4	-	Y	No	Only via MCC codes
2.1.3.1	-	Y	Y	Y
2.1.3.2	-	Only via MCC codes	To be confirmed	Only via MCC codes
2.2.1a1	-	No	Y	No
2.2.1a2	-	Y	Y	No
2.2.1a3	-	Y	No	No
2.2.1a4	-	Y	No	No
2.2.1b1	-	Y	-	Y
2.2.1b2	-	Y	-	Y
2.2.1b3	-	Y	-	Y
2.2.1b4	-	Y	-	Y
2.2.1c1	-	Y	-	Y
2.2.1c2	-	Y	-	Y
2.2.1c3	-	Y	-	Y
2.2.1c4	-	Y	-	Y
2.2.2a1	-	No	No	No
2.2.2a2	-	Y	No	-
2.2.2a3	-	Y	No	No
2.2.2a4	-	Y	No	No
2.2.2b1	-	Y	No	No
2.2.2b2	-	Y	No	Y
2.2.2b3	-	Y	No	Y
2.2.2b4	-	Y	No	Y
2.2.2c1	-	Y	No	No
2.2.2c2	-	Y	No	Y
2.2.2c3	-	Y	No	Y
2.2.2c4		Y	No	Y
2.3 Ind.1	-	Payments with Norwegian cards over the Internet in Norway and abroad, and payments by foreigners' cards over the Internet in Norway (100%)	Aggregated online payments across residents and across residents and non-residents (100%)	-

2.3 Ind.2	-	Country where the card has been issued, and country where the card has been used (100%)	-	-
2.3 Ind.3	-	Merchant Category Codes (100%)	-	-
2.3 Ind.4	-	Use of cards issued abroad on user locations in Norway broken down geographically by country, and for Norway also by postal code (100%)	-	-
3.1	No	Y. "Online FTD transactions" = "Card not present"- transactions in the VISA and MasterCard systems	No	No
3.2	No	Y. Mobile payments" = "Card not present" – combined with "tokenised transactions" in the VISA and MasterCard systems	No	No
3.3	No	Y. Source 1: Highest number of income online payments can be measured for receivers indirectly only -- by MCC-codes. Norwegian virtual user locations may be identified by source 2	No	No
3.4	No	Y	No	No
3.5	No MCC available	We have no information about the nature of payers over the internet today, but we have been informed from source 1 that it should be possible to report the internet use of Norwegian cards held by persons or entities. Receivers can be broken down by MCC codes	-	MCC classification
3.6	Only domestic/foreign card or payment at domestic/foreign terminals	See answer of Q 2-3, indicator 2	-	The level of detail will depend on the number of transactions in order to protect confidential data.
3.7	-	See answer of Q 2-3, indicator 3 – MCC can be linked by NACE to sector. Notice: assuming Sector is the variable that can have the values "Public/Private"	-	MCC classification
3.8	-	MCC-codes are used, but they will not be exact to distinguish between goods and services	-	MCC classification
3.9	-	Quarterly	Yearly	Yearly
3.10	-	Y. Source 2: Individual transactions >=NOK 25 000 (Not yet, but from 2020: DSOP-data)	No	No
3.11	-	Y. Information in general about type of transaction and times of settlement /settlement frequency	No	Y. Data could be made available on a daily basis.
3.12	-	Procedures controlling the validity of data and metadata and completion of the reporting	For the time being we have no idea how to check FTD for accuracy and representativeness.	Data from various sources will be systematically compared to check for consistency.

Appendix 11. Problems with sharing platforms

Online platforms are important elements in the operational definition as they are a data source for shared economy linked transactions. The Study to Monitor the Economic Development of the Collaborative Economy at Sector Level in the 28 EU Member States (Febr. 23, 2018) provides a useful list of SEP. Bulgaria has six platform in the transport sector. Two of these platforms ceased to operate.

One can find many sharing platforms being closed after paper publishing. One interesting example is solar sharing network Yeloha (Codagnone & Martens 2016). Yeloha started in 2012 to harness knowledge, data, and enthusiasm of individuals decided to "go solar" and build up a network of roof solar energy production and consumption. Some of roofs are sunny and produce a lot of electricity. Some are not sunny and families will never invest on a solar system. However, both modern technologies and regulative framework allow for sharing unused electricity produced on sunny roofs with network members having cloudy roofs. This is, in a nutshell, what service Yeloha platform provided until 2016. Co-founder of Yeloha Amit Rosner explains what happen in 2016:

"Eventually Yeloha shut down because we could not raise the financing we needed in order to massively grow our network. Timing hurt. The so called "Venture Capital winter" of 2016 coincided with the turmoil in the solar stock market and the bankruptcy of multi-billion-dollar SunEdison, venture investors fled from solar, and strategic investors crystallised their strategy around profitability."

Yeloha case demonstrates a sharing platform could evolve in different ways over time because of combination of factors: economic and legal environment, venture capitalist expectations, etc. May be statistics about sharing platform demography (birth rate, death rate, average expected life of sharing platforms across branches) could provide insights on how sharing economy evolve over time, do platforms stick to the idea of promoting sharing economy or morph into business as usual, or cease operation.

Another problem with platforms is the challenges of separating SEP data on sharing economy from the SEP data of business as usual, since many SEPs contain both types of data. A research shows nearly half of Airbnb revenues are accrued to hosts with multiple listing in New York (Tom Slee 2014 available at https://www.jacobinmag.com/2014/01/sharing-and-caring/). Multiple listing has nothing to do with sharing economy. Such a concentration of revenues could be found in other cities as well. Multiple listings suggest it is not realistic to accrue all revenues of Airbnb platform to SE. We should have information about rentals available round the year and rentals available for a couple of months. Most likely the former is not sharing, and latter are sharing. Unfortunately, experience suggest Airbnb is not ready to share information about transactions in the long run.

Hopefully, sharing platform will turn more cooperative in the future. There is a strong tendency toward imposing regulation on sharing platforms. Two examples. By the end of 2018 the UK Court of Appeal ruled that the Uber drivers are to be considered workers rather than independent contractors. Being workers, they are eligible for holiday pay, sick pay, and other workers benefit. As independent contractors they are not eligible for such benefits. In 2016 a US District Court confirmed the ability of city administration of San Francisco to hold Airbnb liable when hosts operate illegal rentals. City administration can impose fines if Airbnb charges for rentals that are not registered with the city. The regulations of sharing platforms is a precondition for having more data, including administrative data sources useful to separate sharing from business as usual